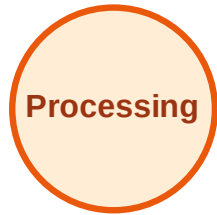


BFS Traversal

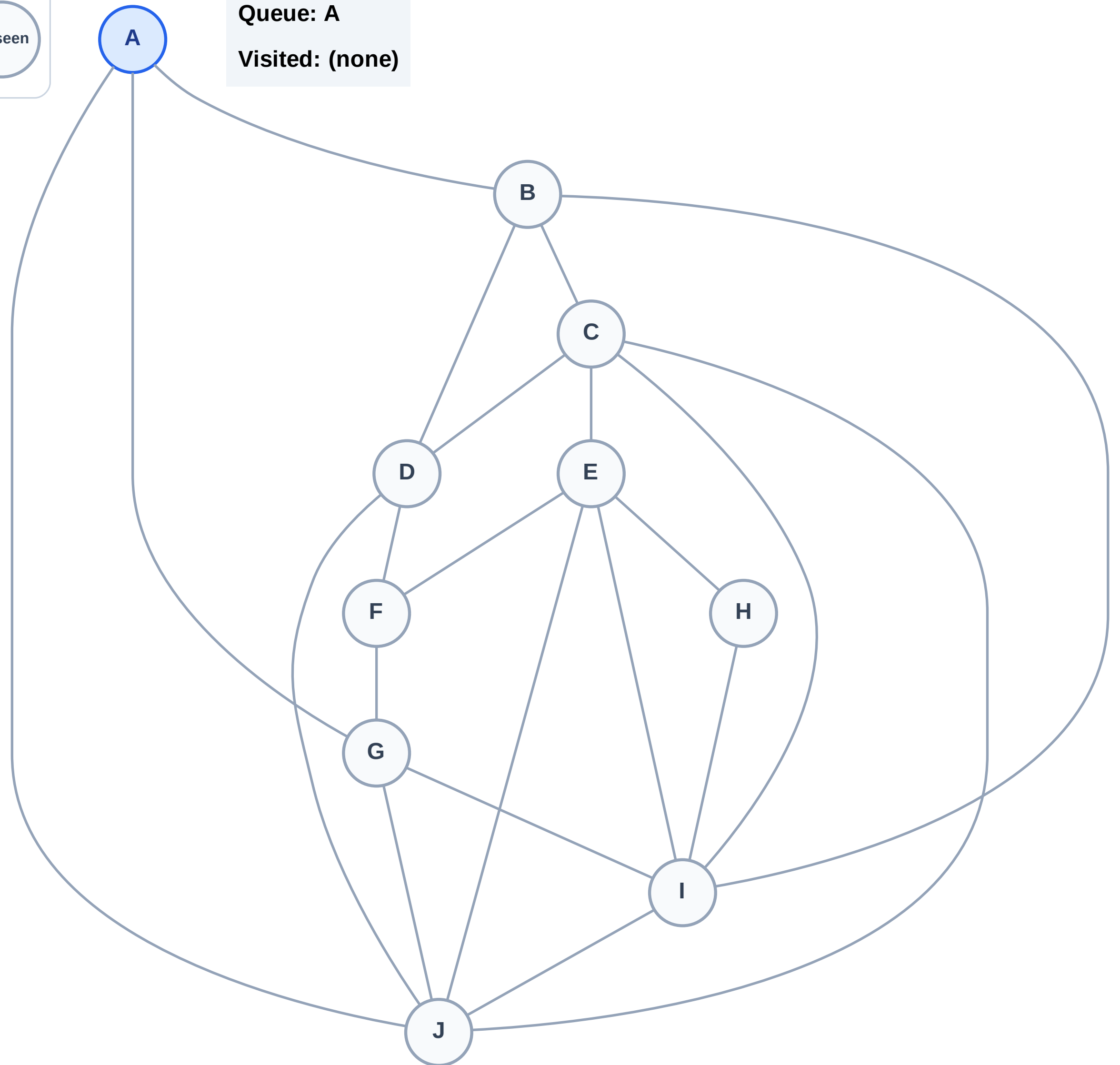
Step 1: Start at A

Legend



Queue: A

Visited: (none)



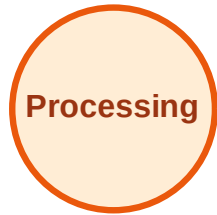
BFS Traversal

Step 2: Process node A

Legend



Complete



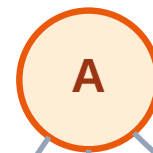
Processing



Discovered



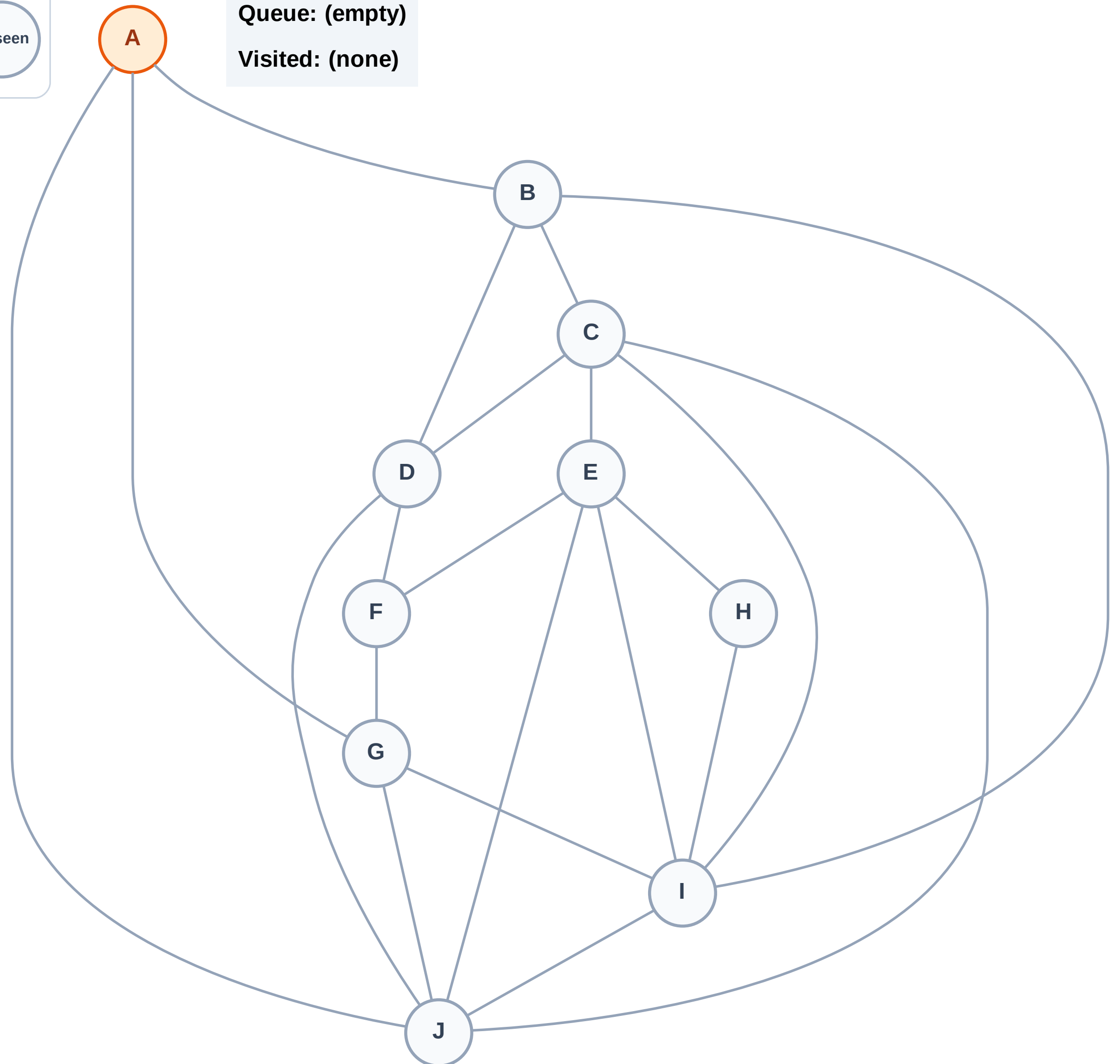
Unseen



A

Queue: (empty)

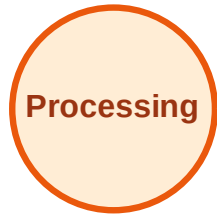
Visited: (none)



BFS Traversal

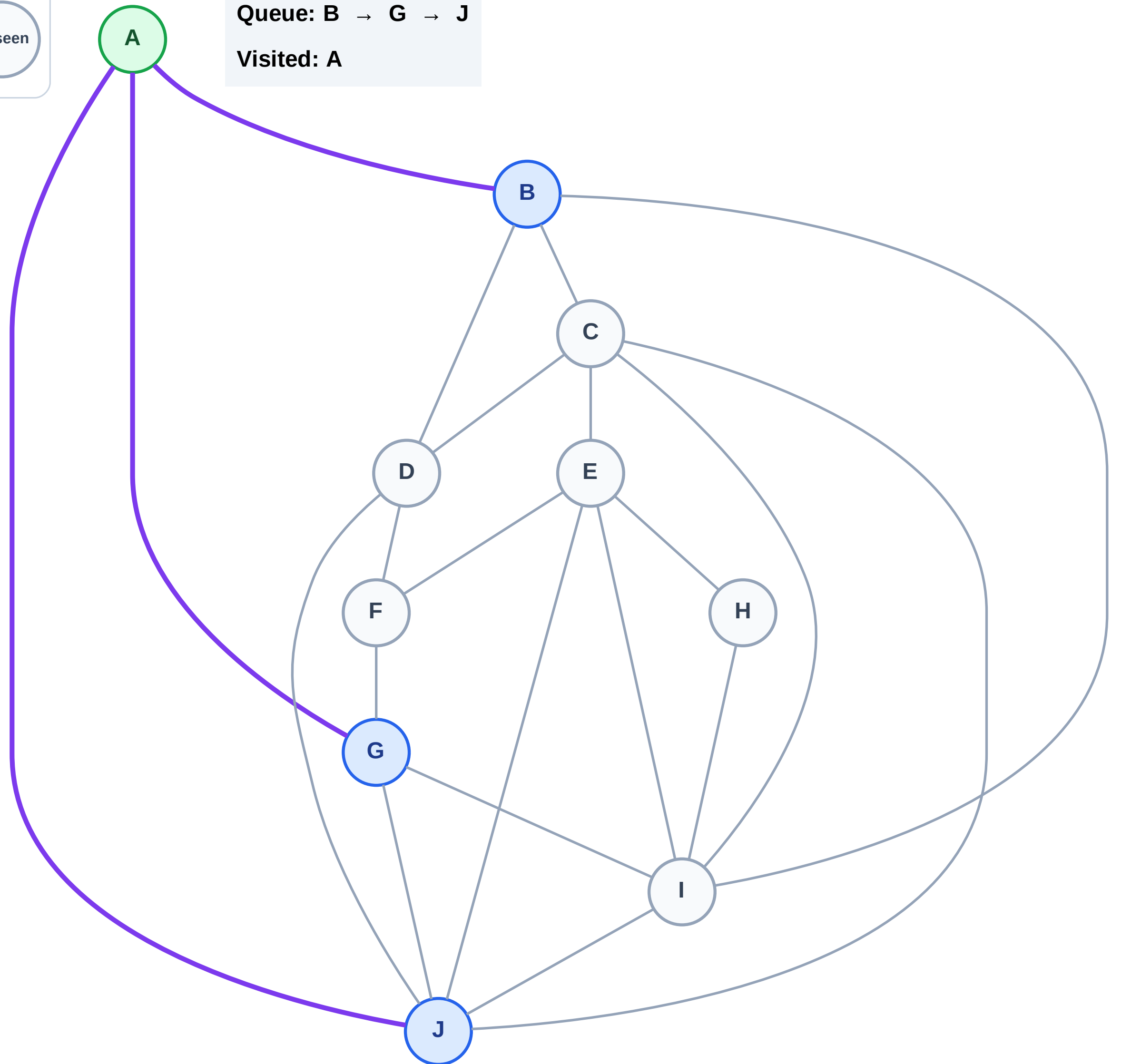
Step 3: Discover B, G, J from A

Legend



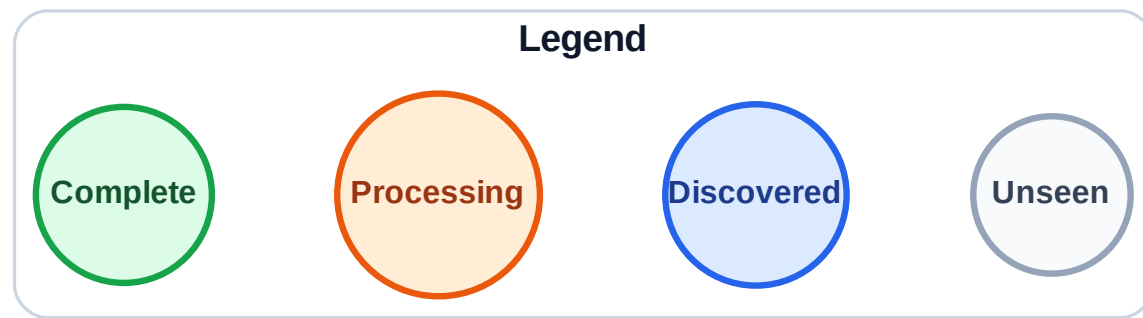
Queue: B → G → J

Visited: A

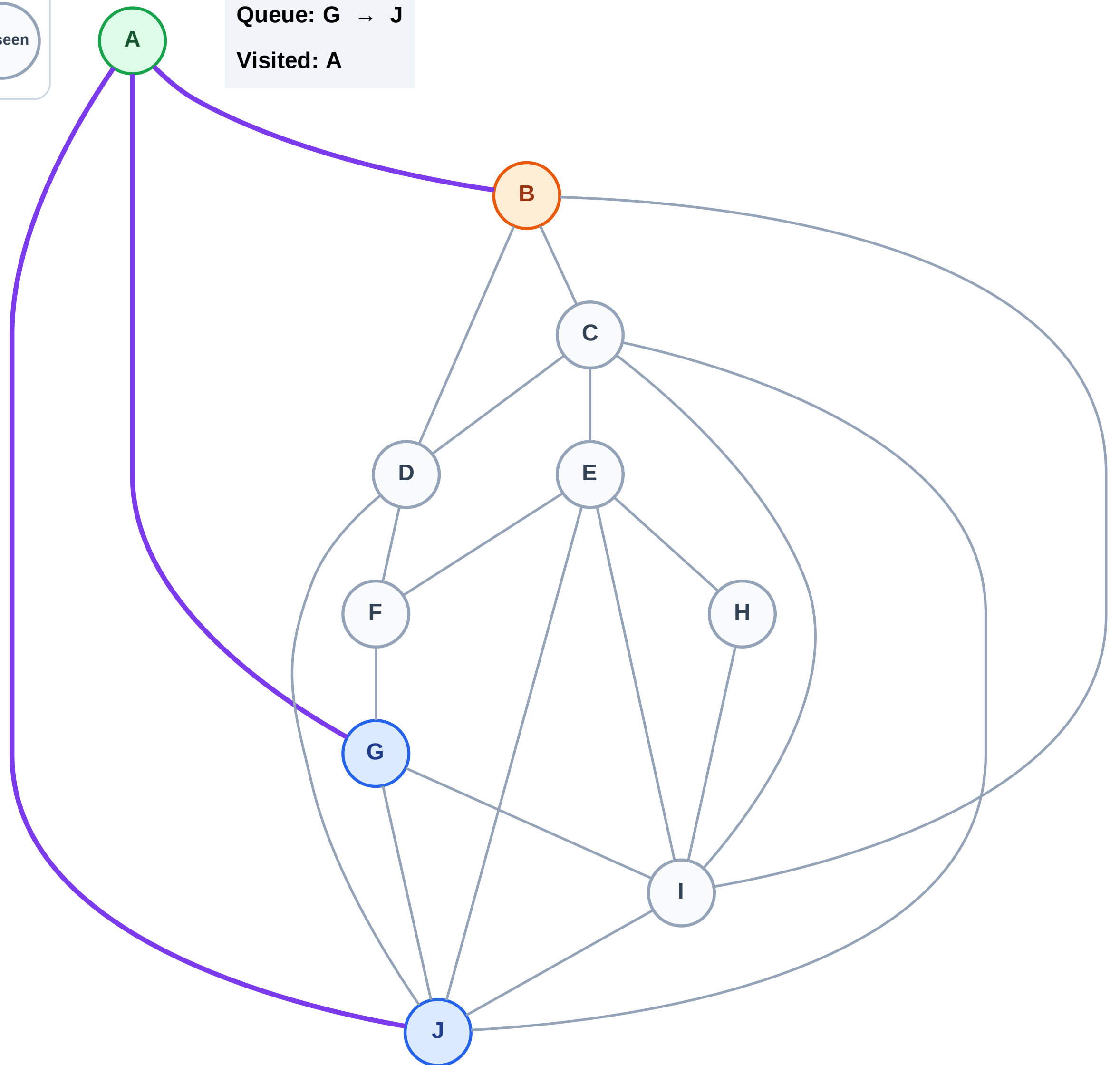


BFS Traversal

Step 4: Process node B



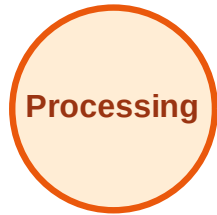
Queue: G → J
Visited: A



BFS Traversal

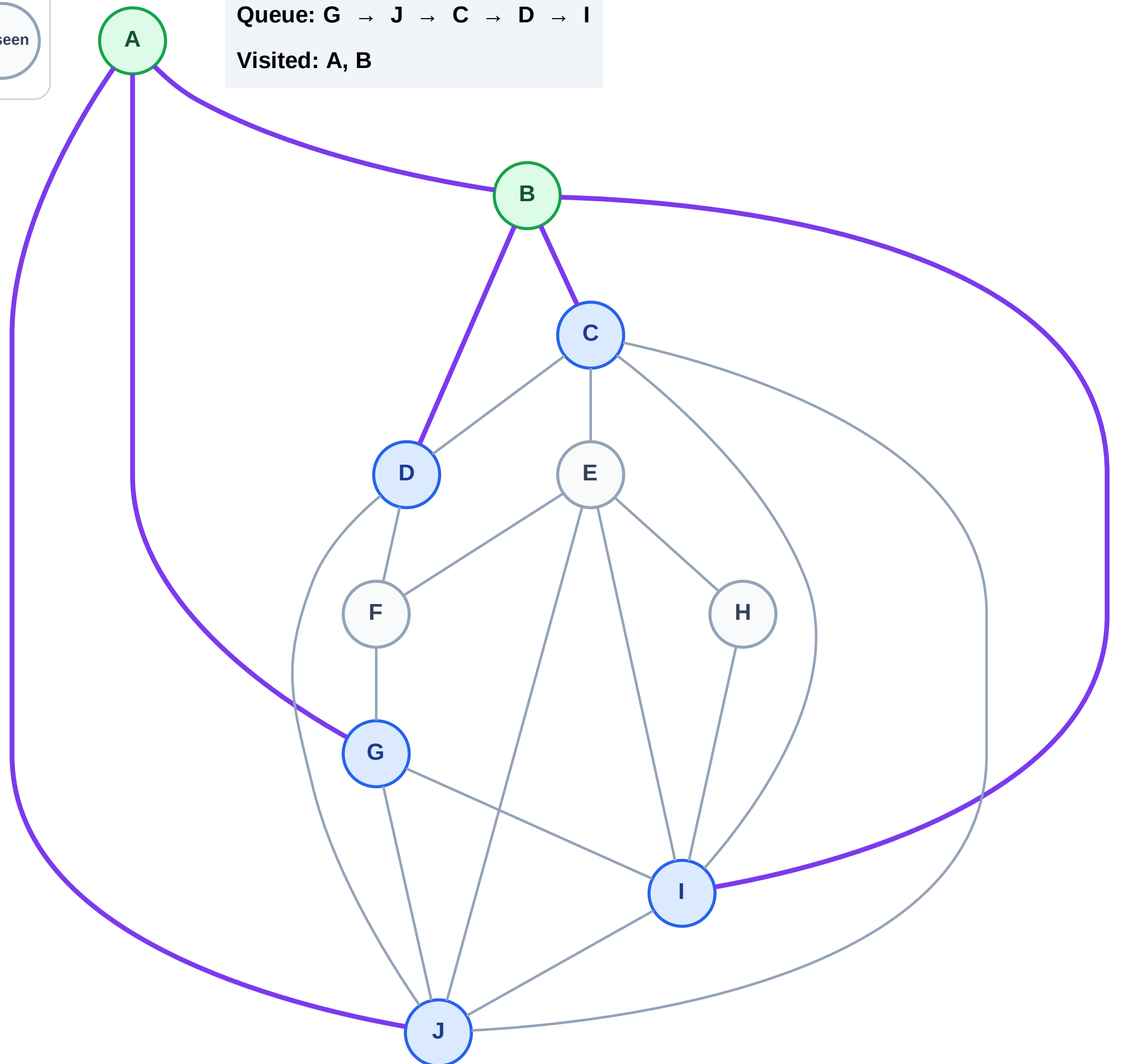
Step 5: Discover C, D, I from B

Legend



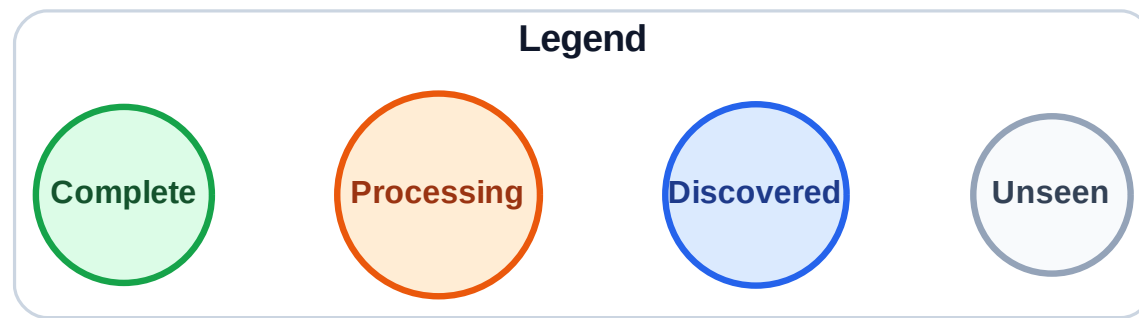
Queue: G → J → C → D → I

Visited: A, B

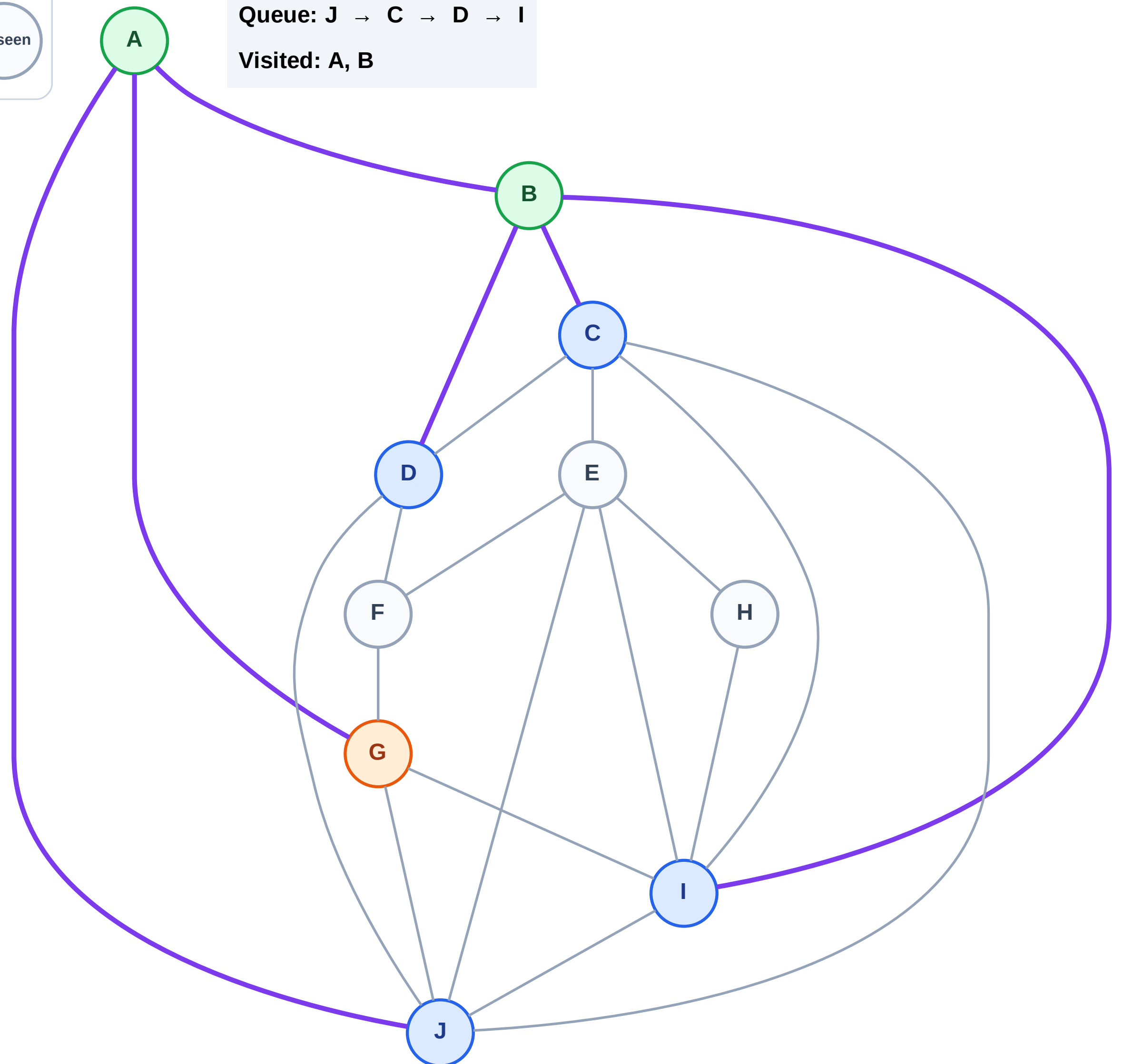


BFS Traversal

Step 6: Process node G



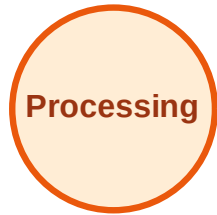
Queue: J → C → D → I
Visited: A, B



BFS Traversal

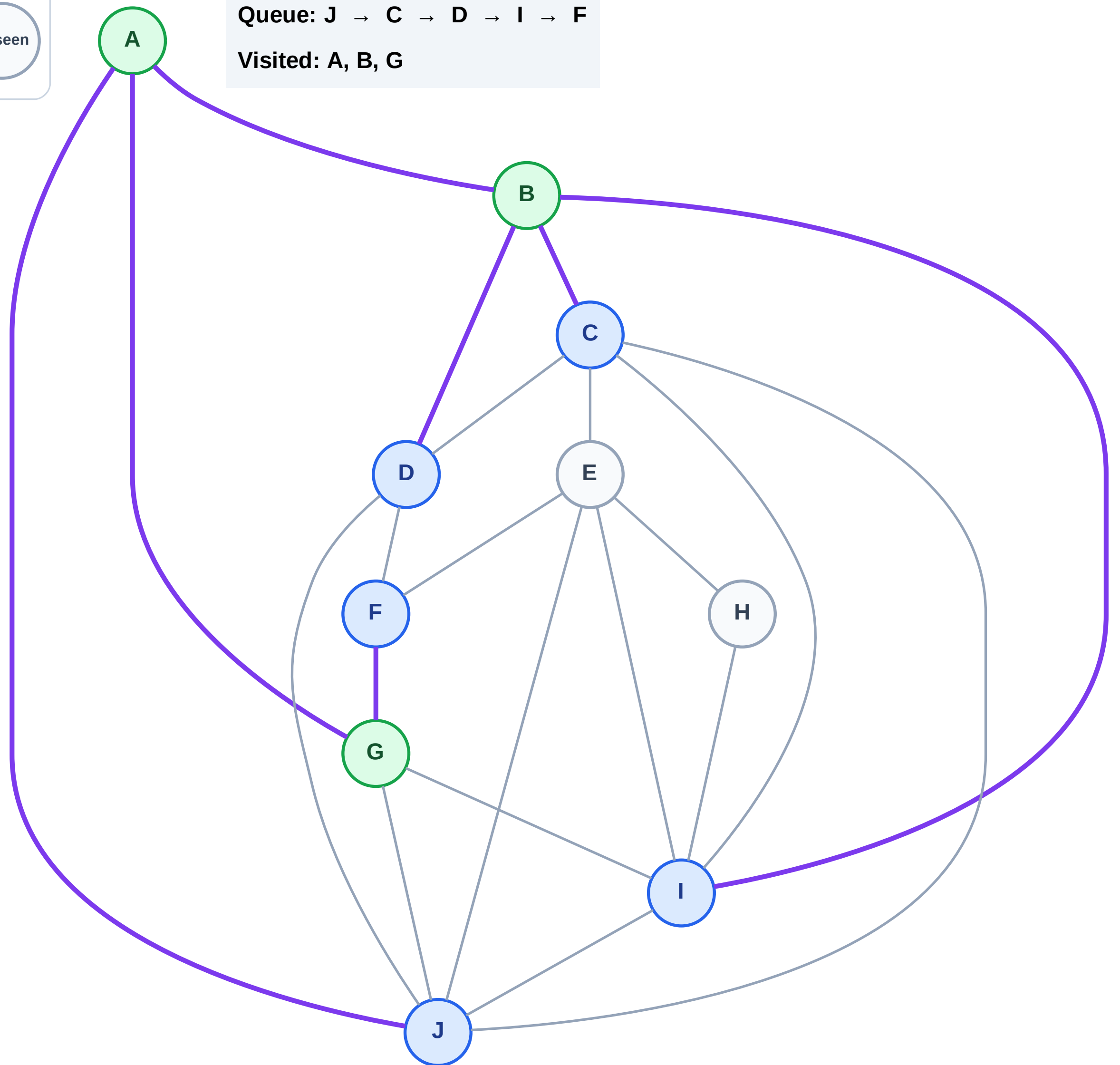
Step 7: Discover F from G

Legend



Queue: J → C → D → I → F

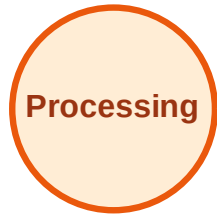
Visited: A, B, G



BFS Traversal

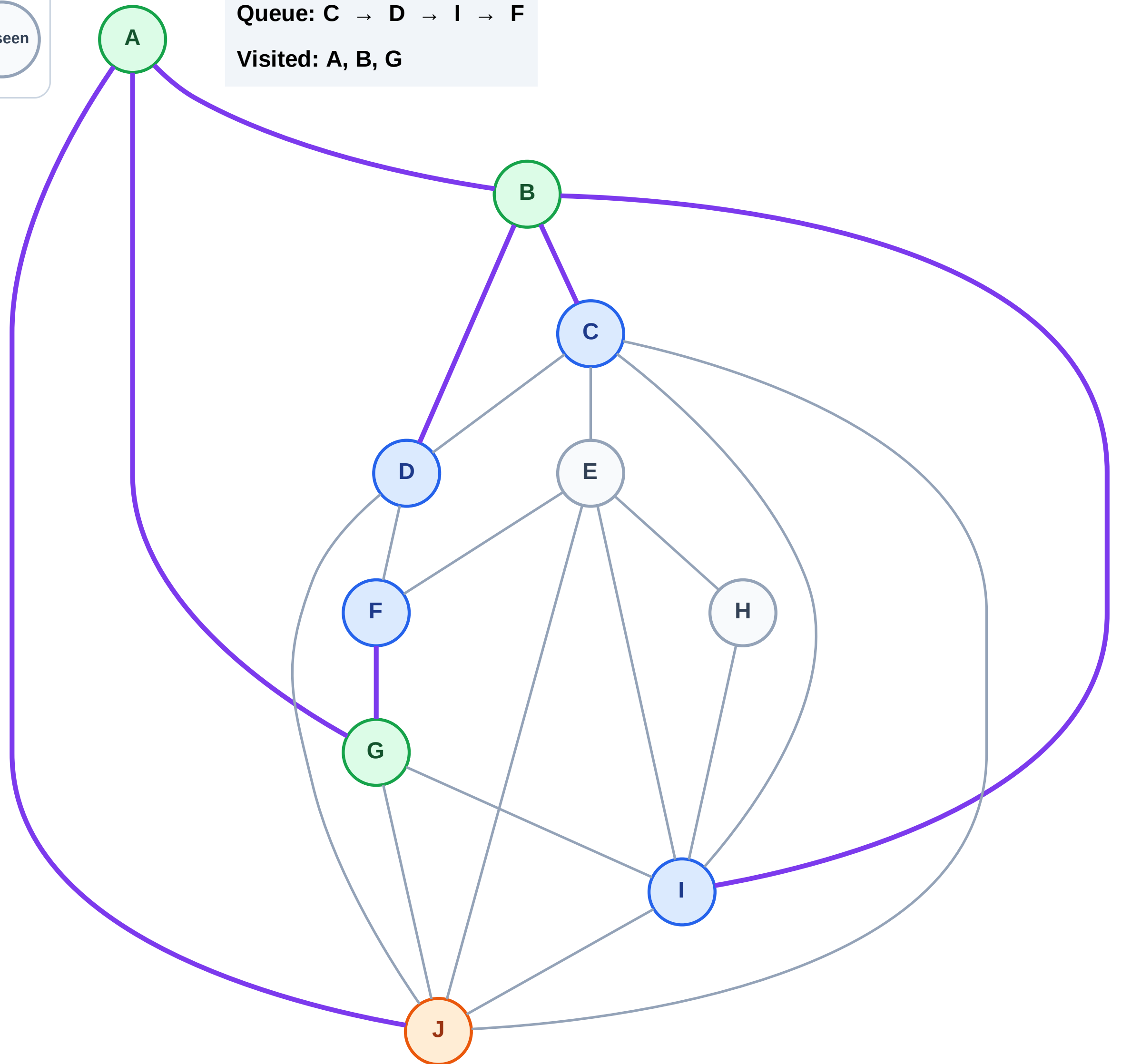
Step 8: Process node J

Legend



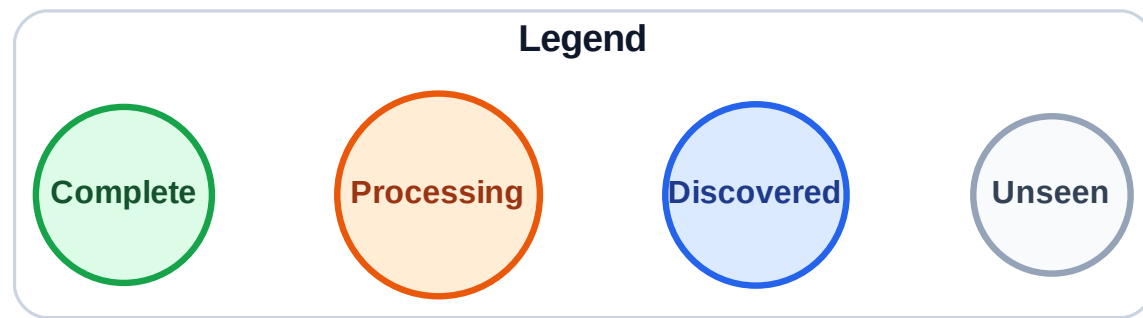
Queue: C → D → I → F

Visited: A, B, G

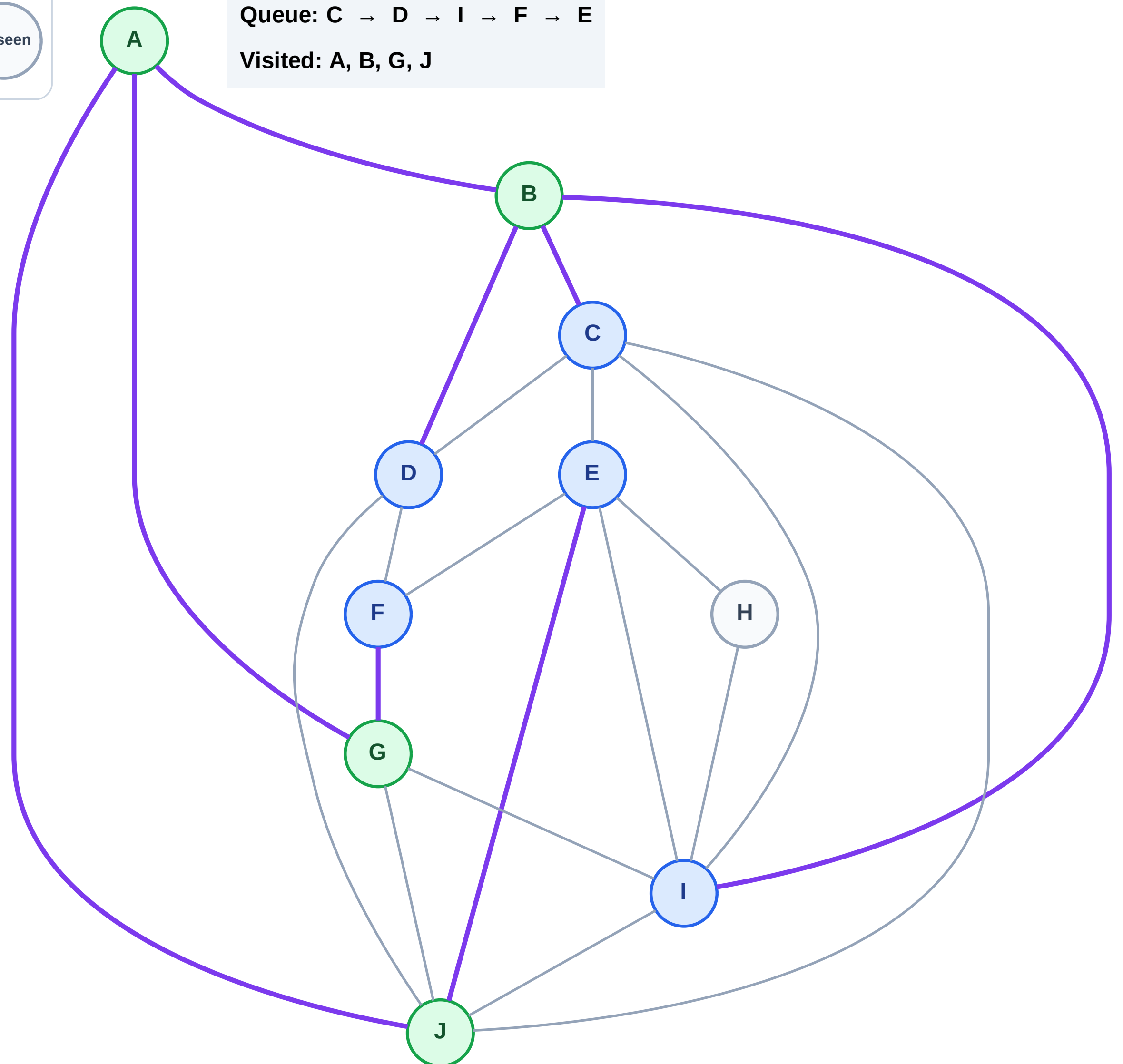


BFS Traversal

Step 9: Discover E from J

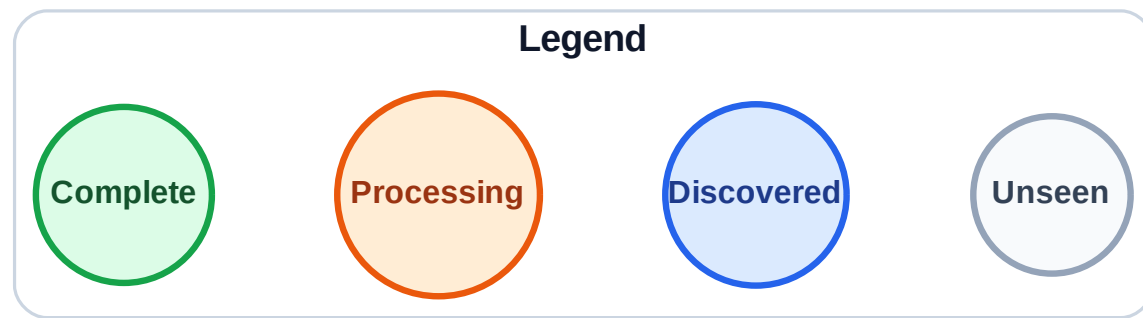


Queue: C → D → I → F → E
Visited: A, B, G, J

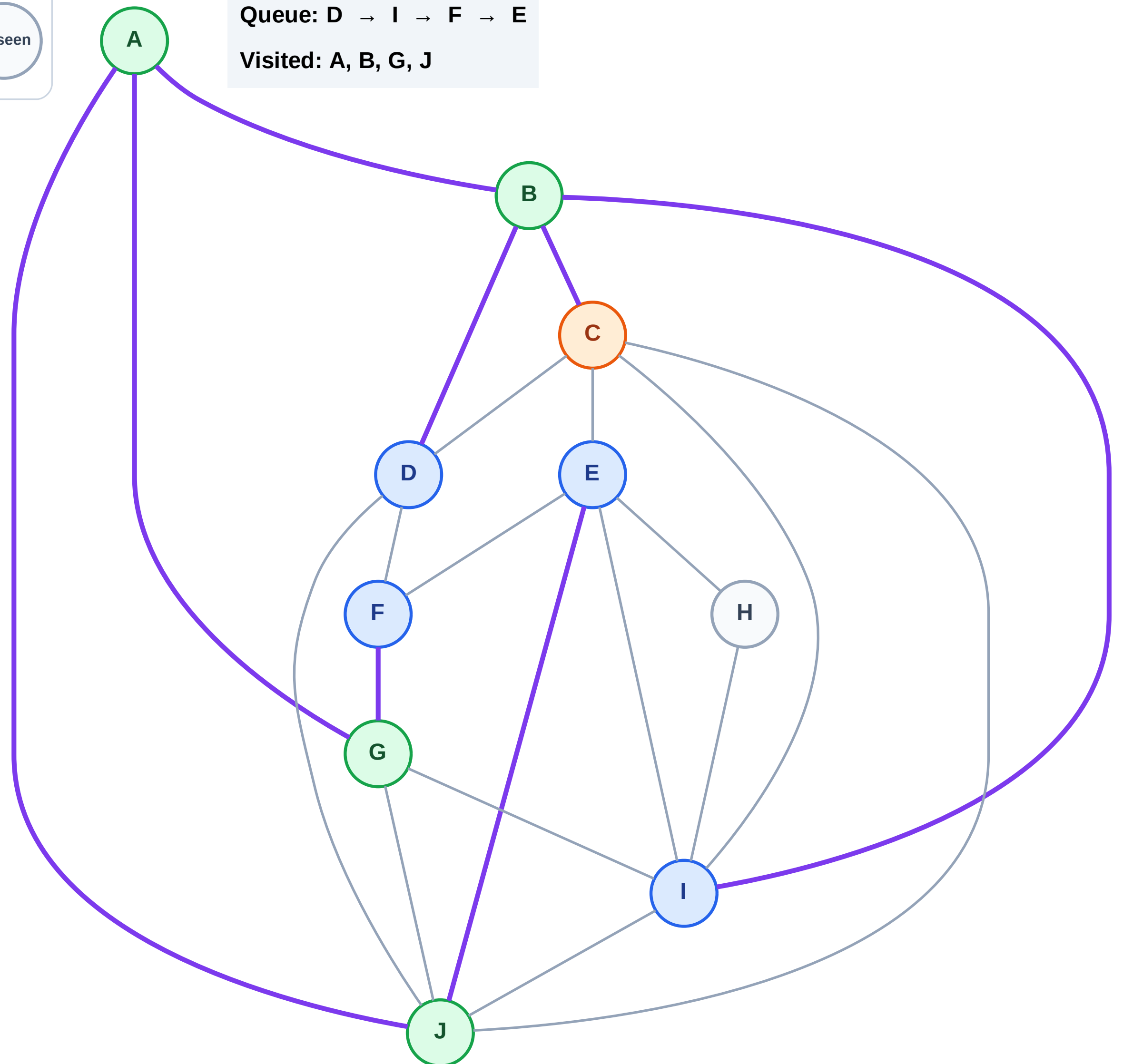


BFS Traversal

Step 10: Process node C



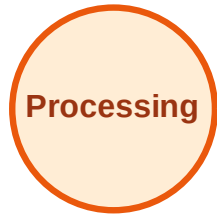
Queue: D → I → F → E
Visited: A, B, G, J



BFS Traversal

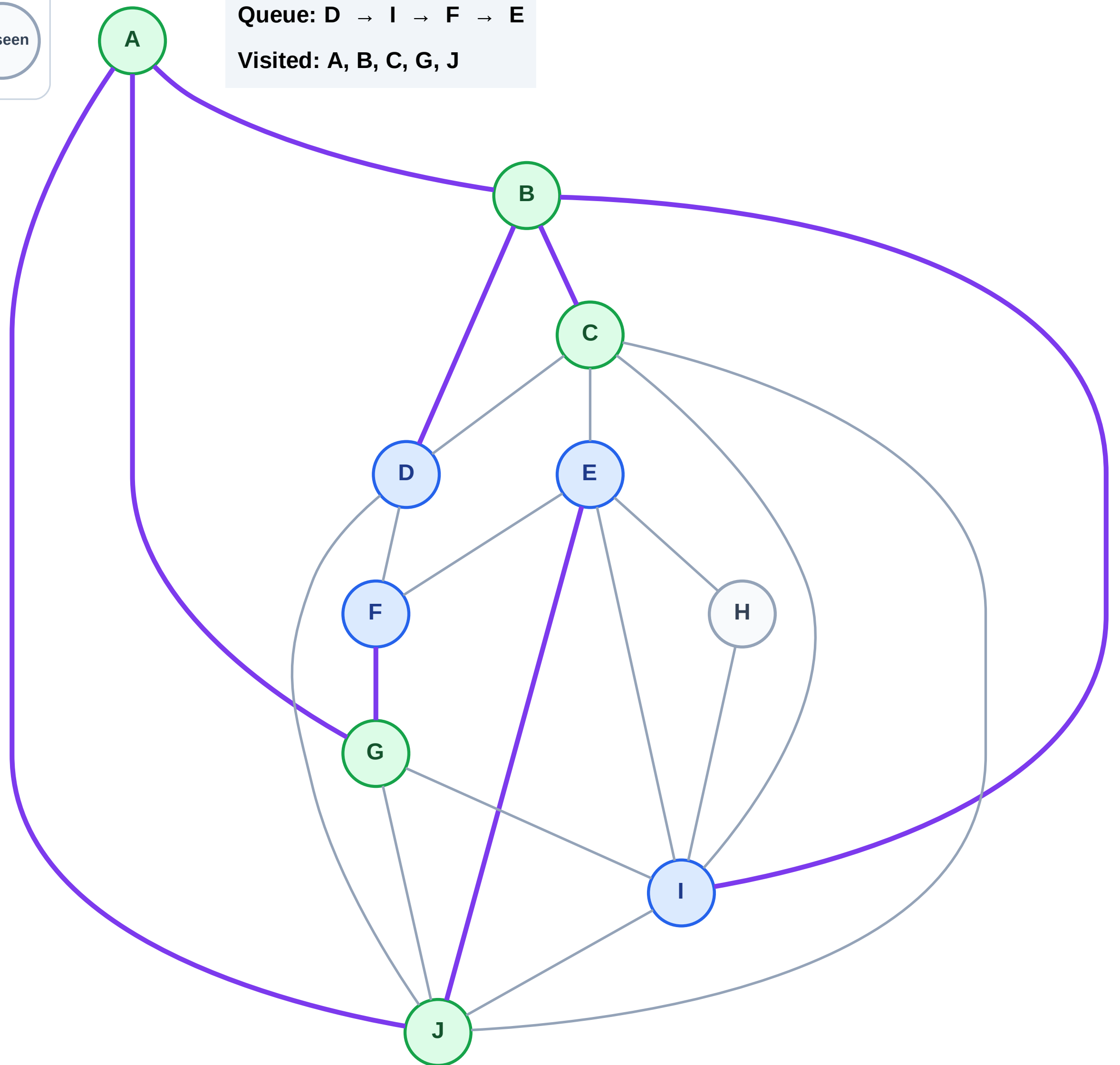
Step 11: No new neighbors from C

Legend



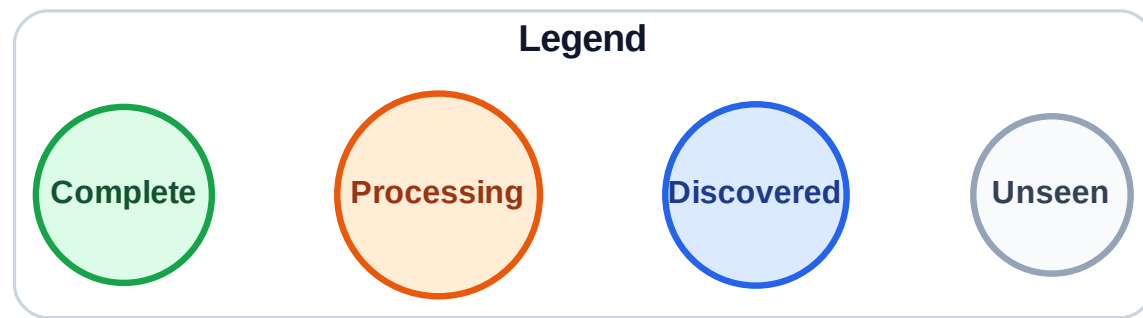
Queue: D → I → F → E

Visited: A, B, C, G, J

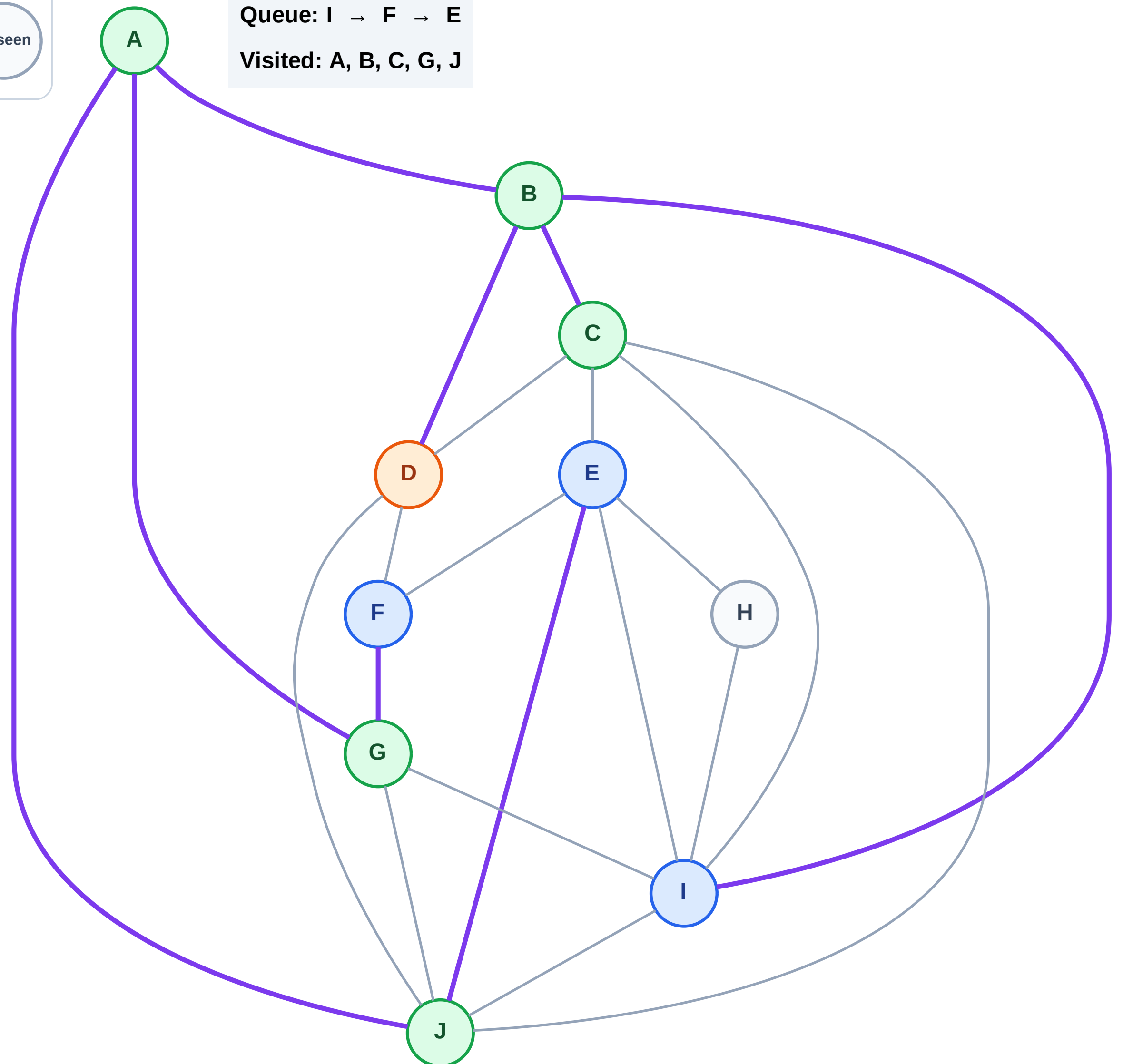


BFS Traversal

Step 12: Process node D



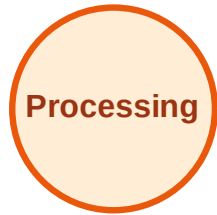
Queue: I → F → E
Visited: A, B, C, G, J



BFS Traversal

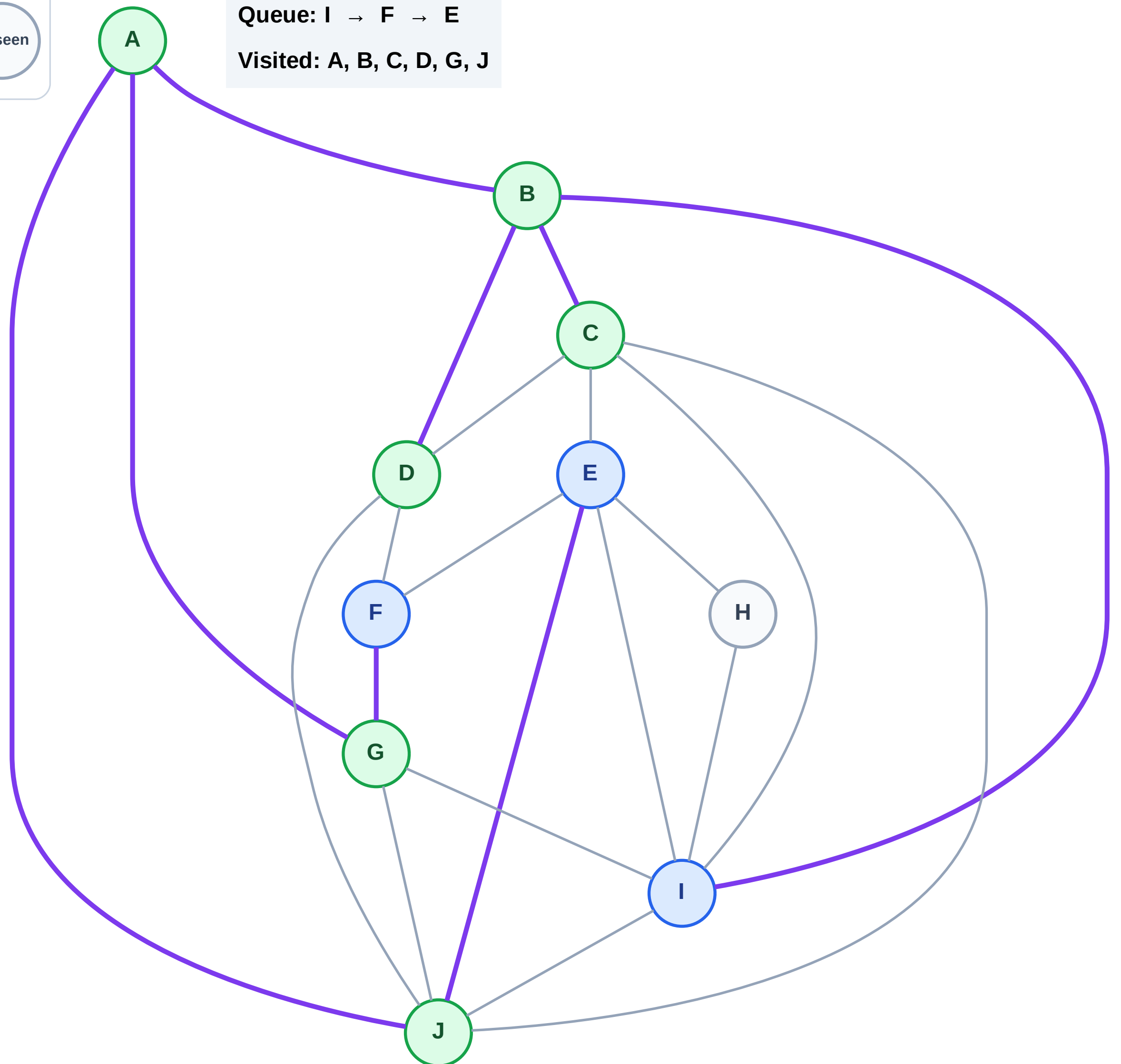
Step 13: No new neighbors from D

Legend



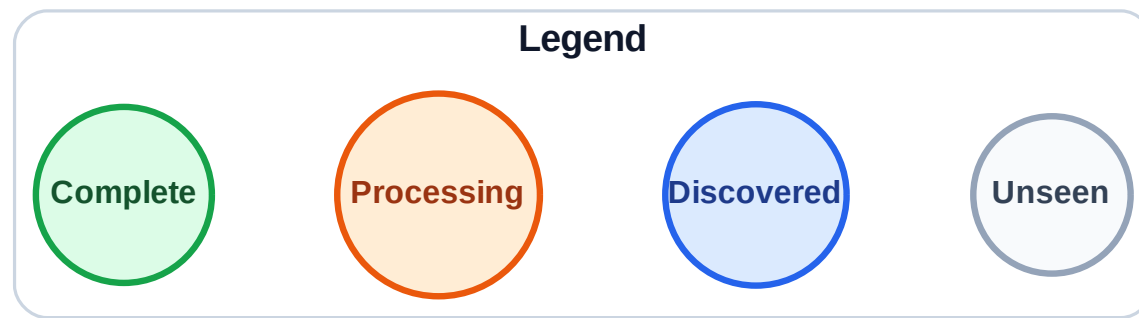
Queue: I → F → E

Visited: A, B, C, D, G, J

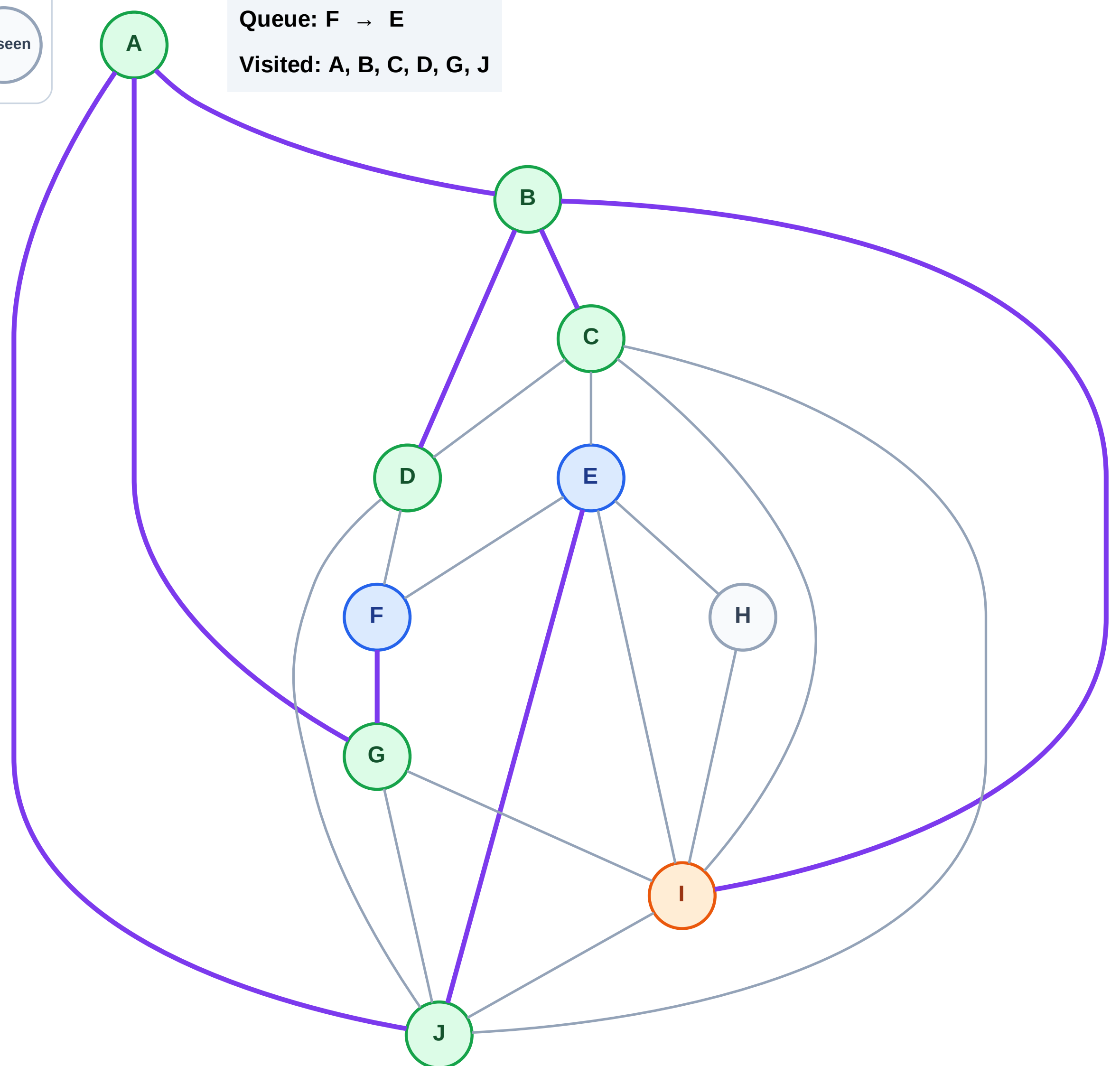


BFS Traversal

Step 14: Process node I



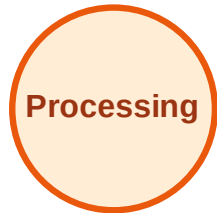
Queue: F → E
Visited: A, B, C, D, G, J



BFS Traversal

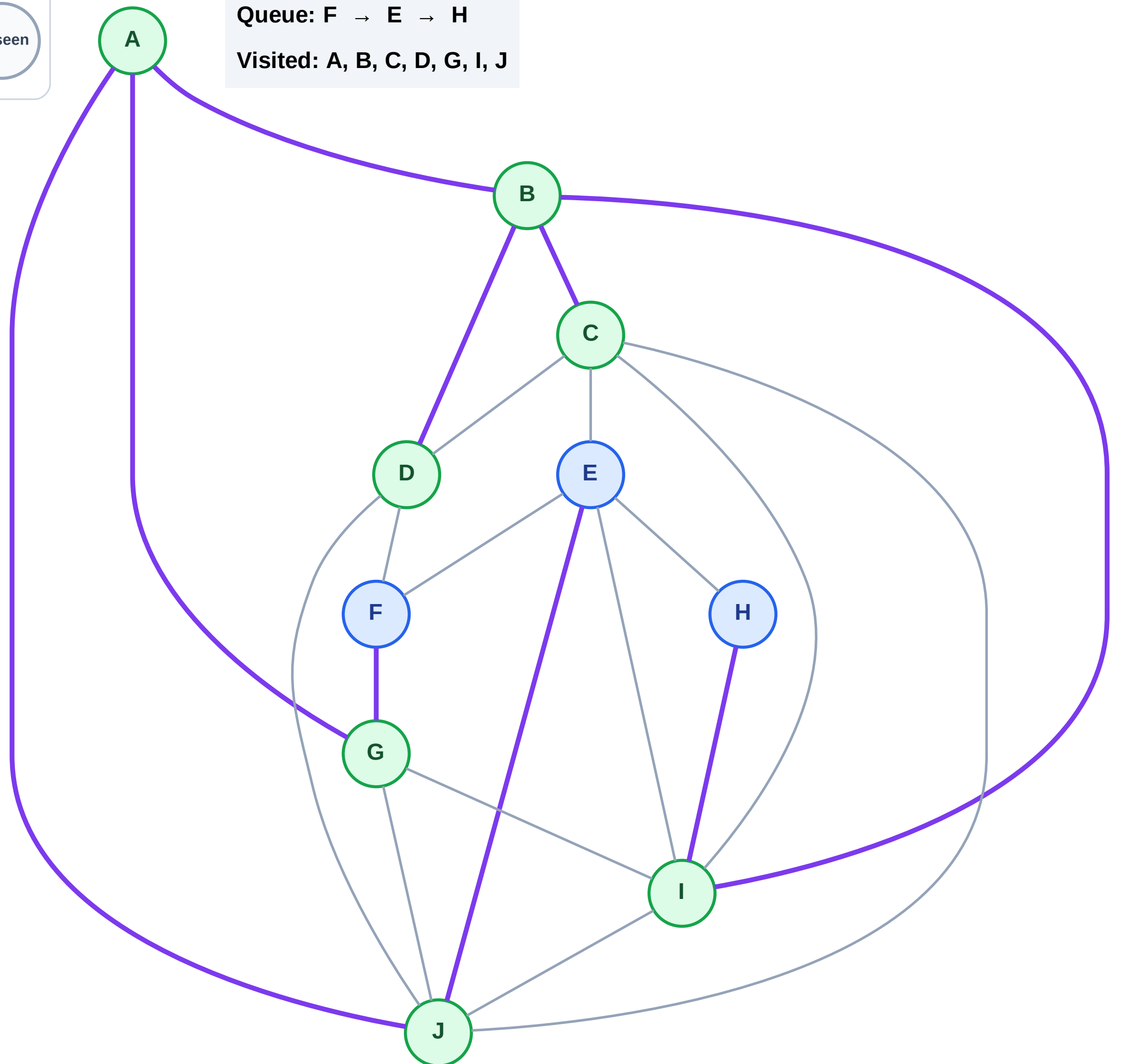
Step 15: Discover H from I

Legend



Queue: F → E → H

Visited: A, B, C, D, G, I, J



BFS Traversal

Step 16: Process node F

Legend

Complete

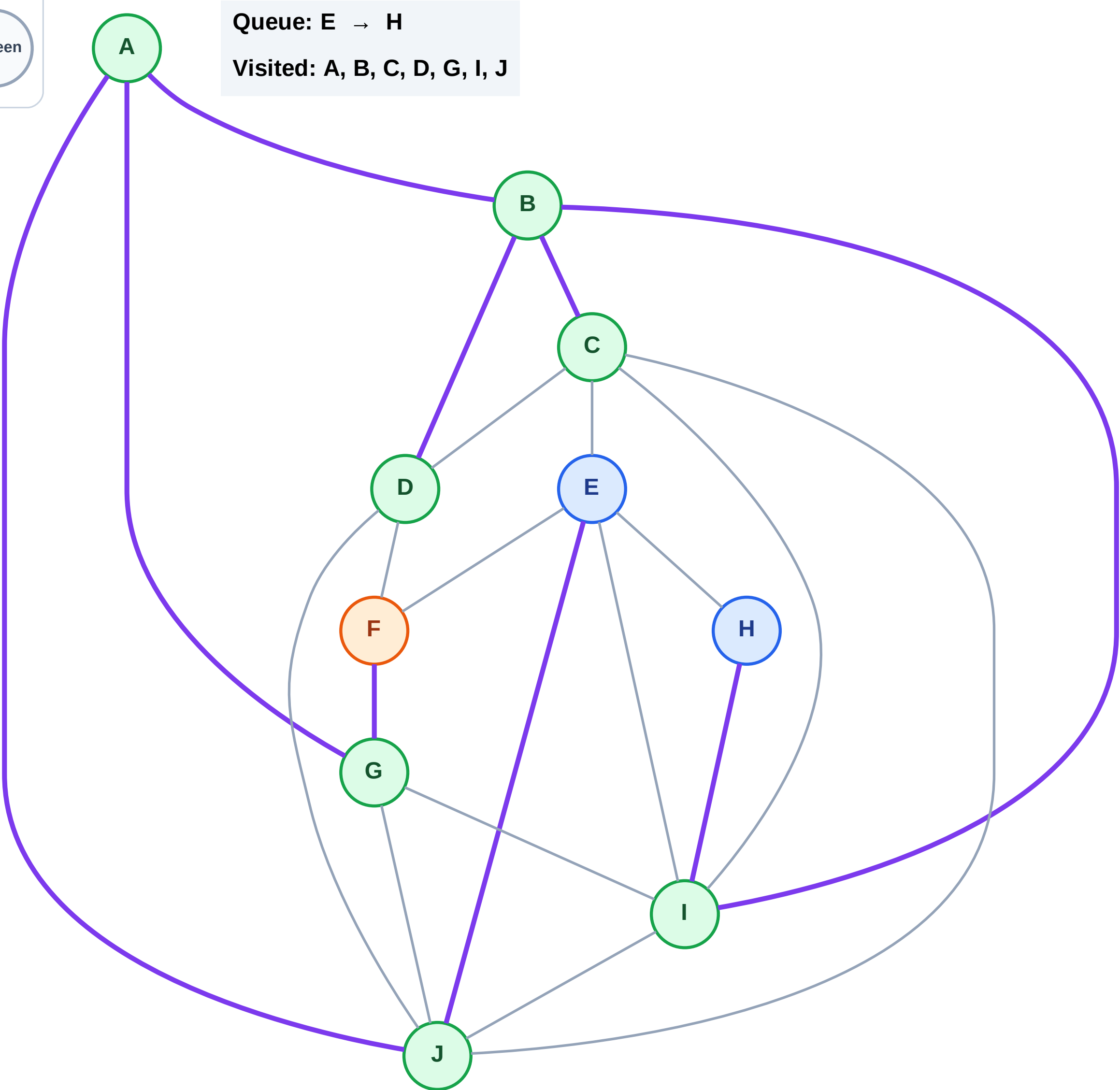
Processing

Discovered

Unseen

Queue: E → H

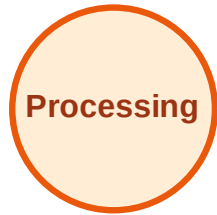
Visited: A, B, C, D, G, I, J



BFS Traversal

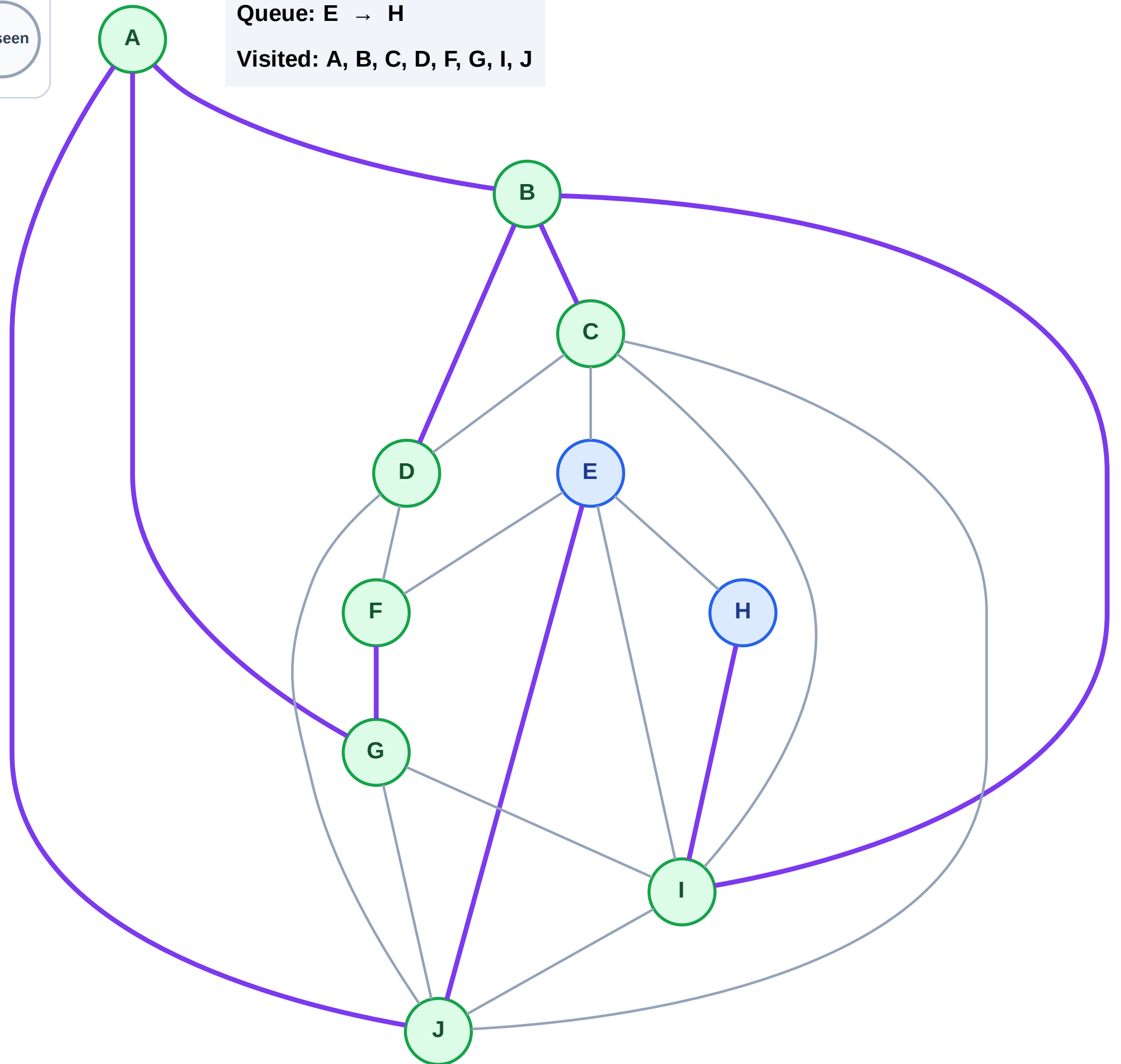
Step 17: No new neighbors from F

Legend



Queue: E → H

Visited: A, B, C, D, F, G, I, J



BFS Traversal

Step 18: Process node E

Legend

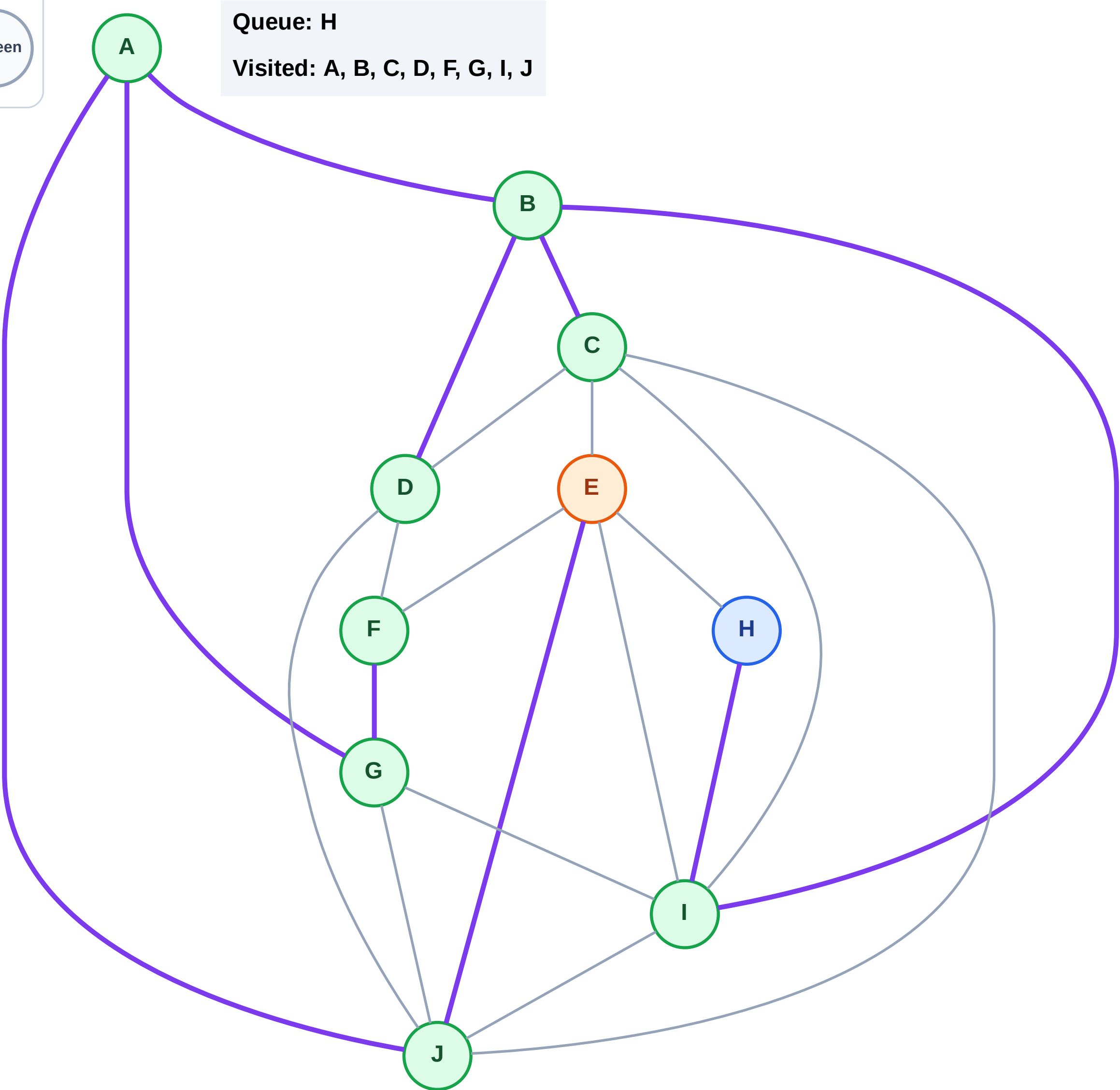
Complete

Processing

Discovered

Unseen

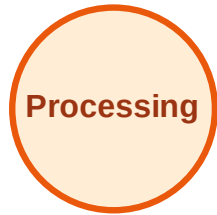
Queue: H
Visited: A, B, C, D, F, G, I, J



BFS Traversal

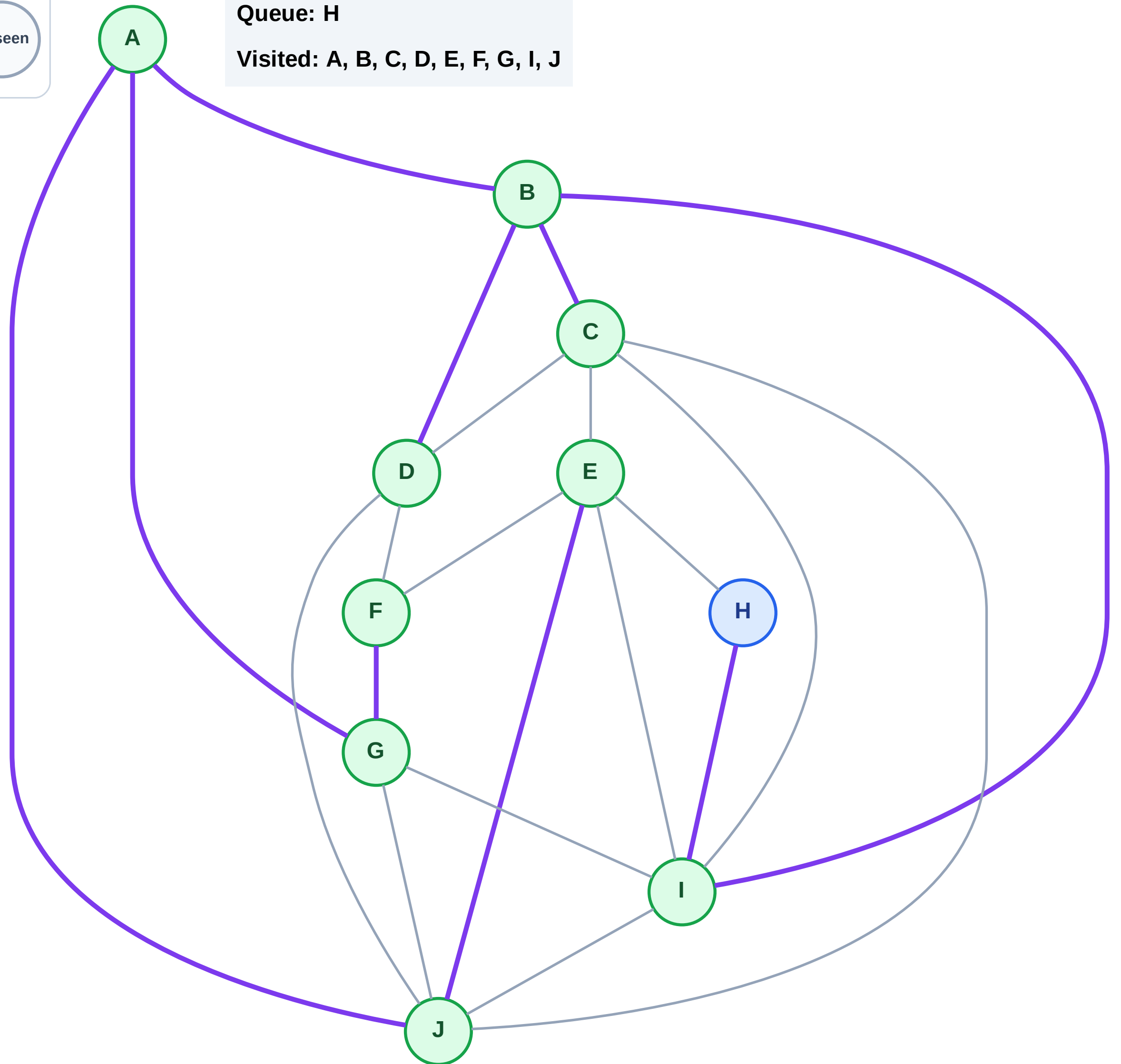
Step 19: No new neighbors from E

Legend



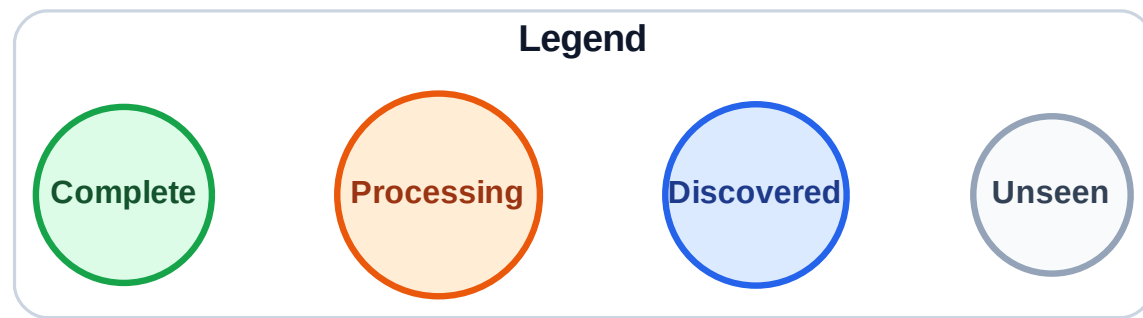
Queue: H

Visited: A, B, C, D, E, F, G, I, J

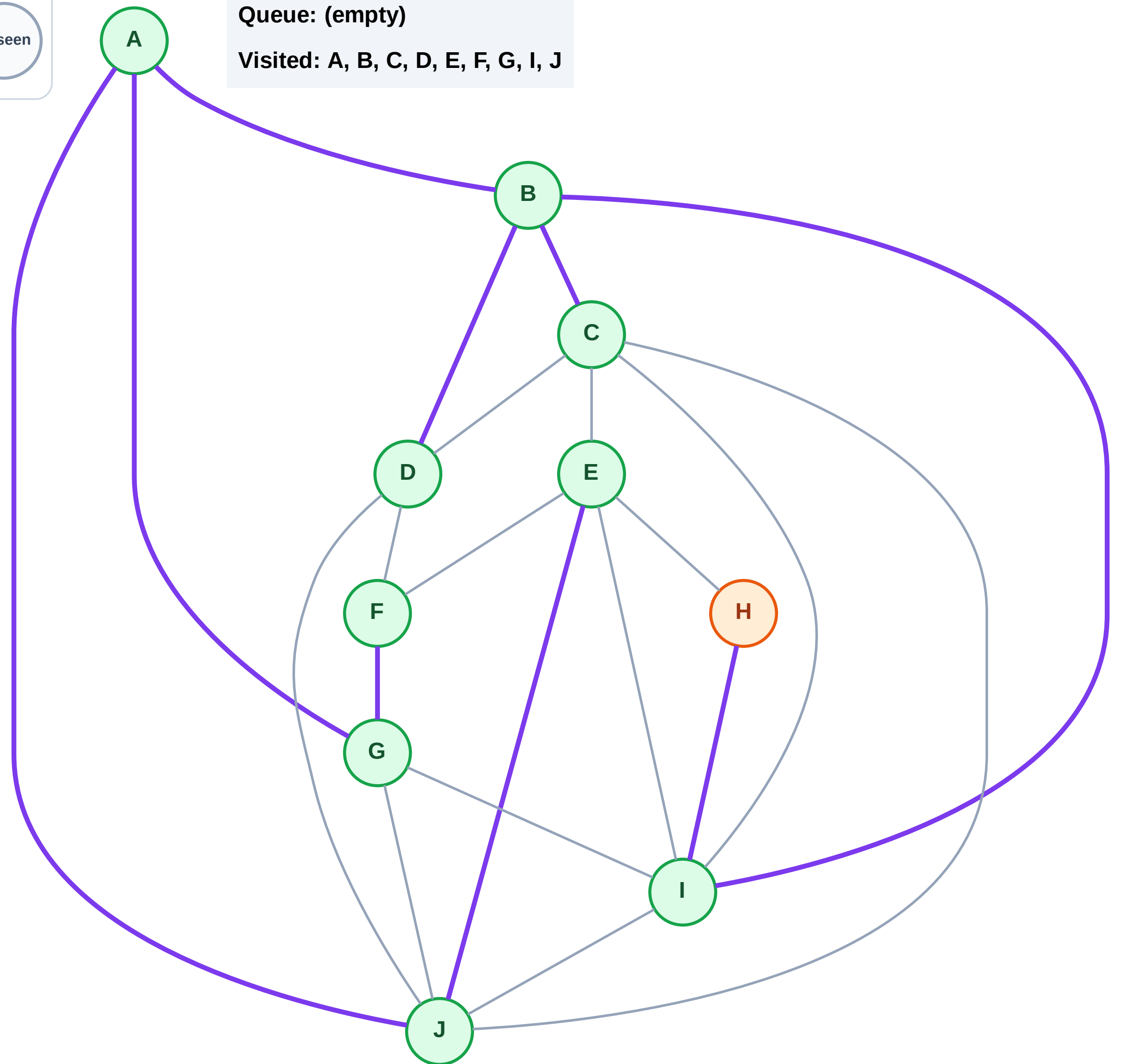


BFS Traversal

Step 20: Process node H



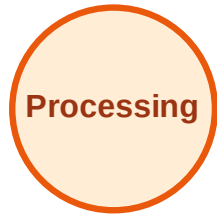
Queue: (empty)
Visited: A, B, C, D, E, F, G, I, J



BFS Traversal

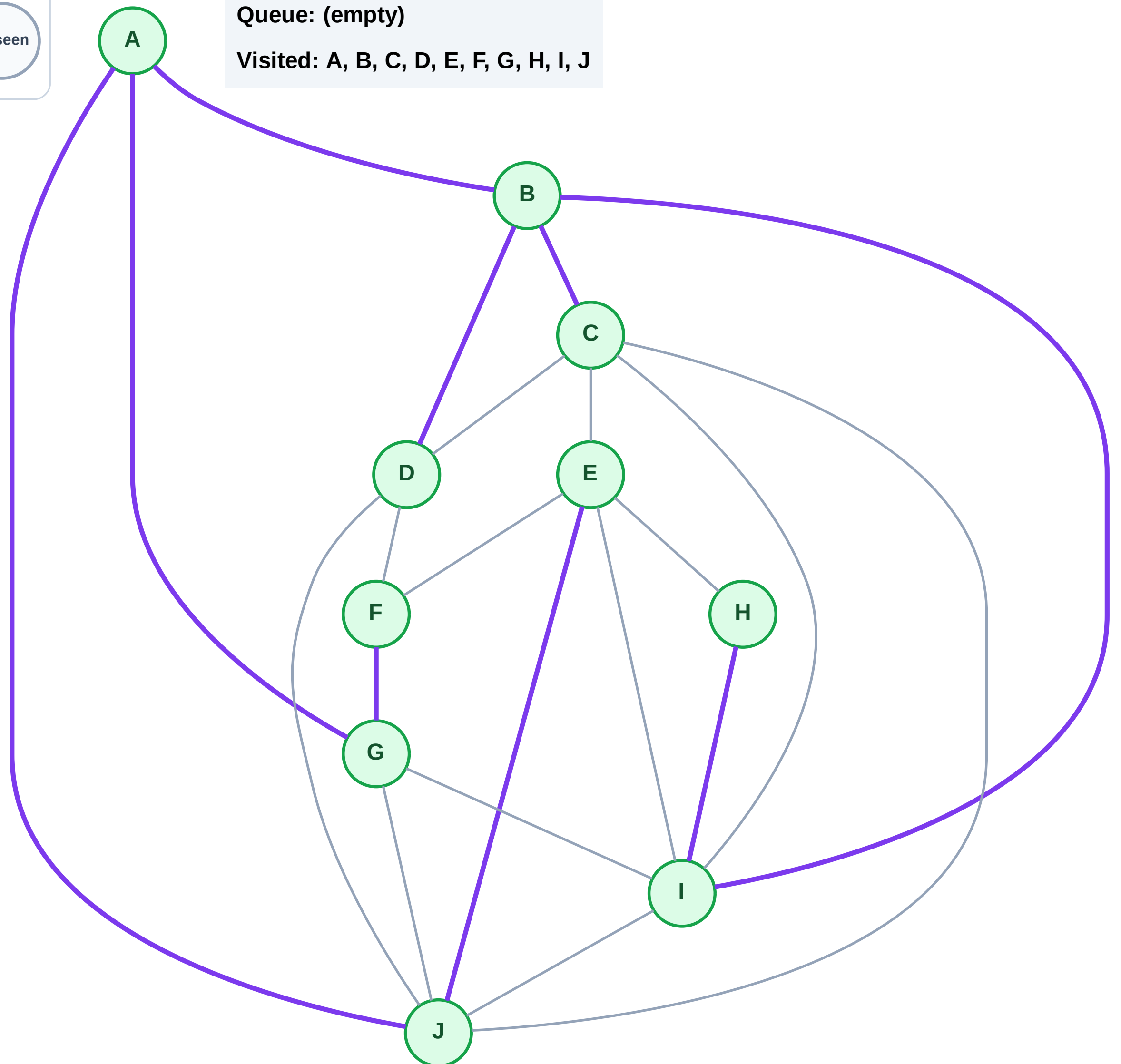
Step 21: No new neighbors from H

Legend



Queue: (empty)

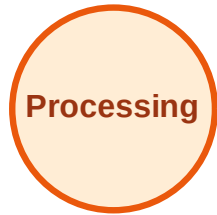
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

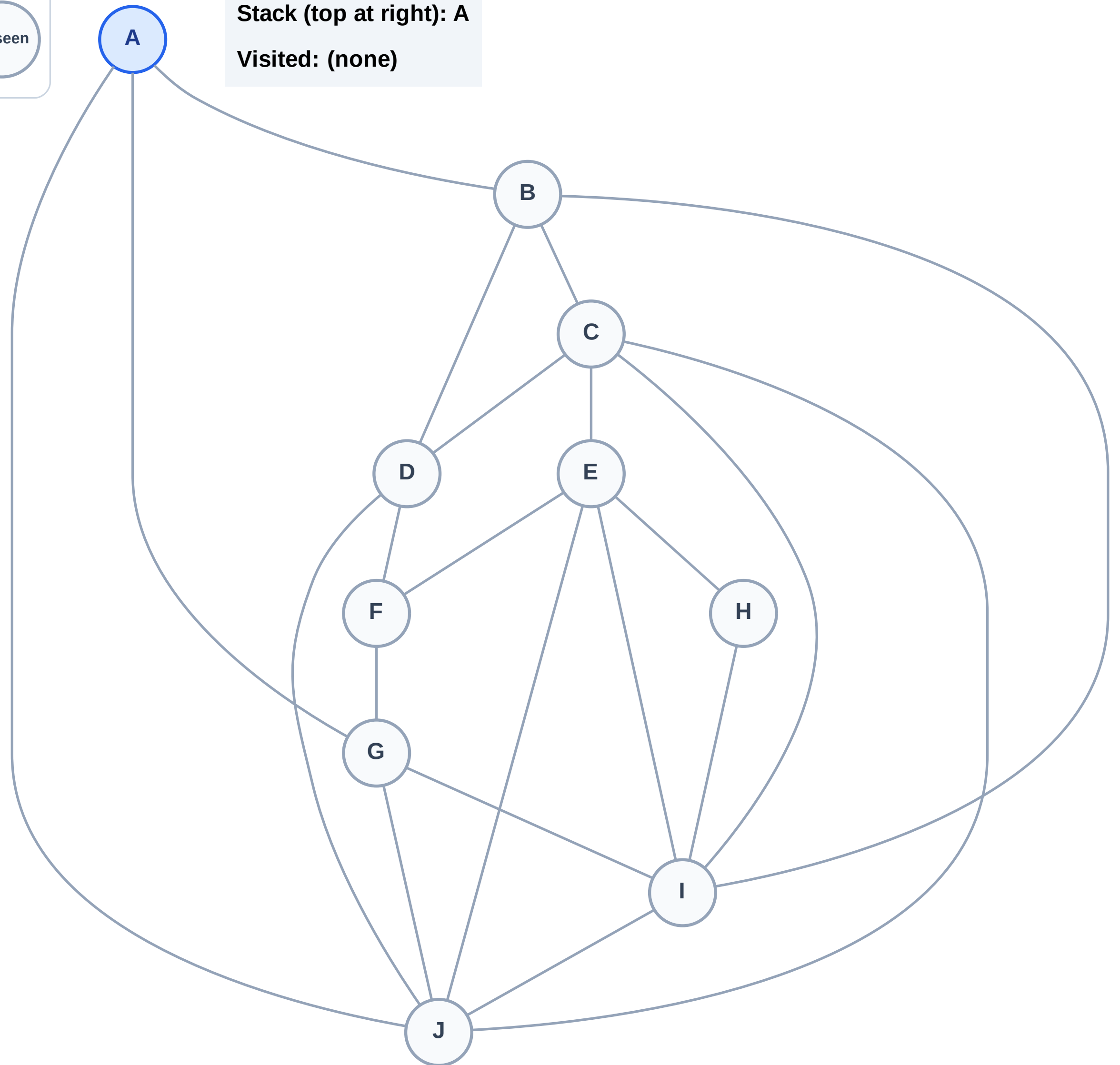
Step 1: Start at A

Legend



Stack (top at right): A

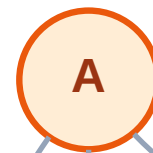
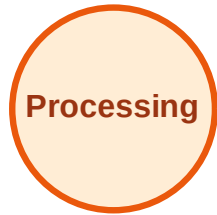
Visited: (none)



DFS Traversal

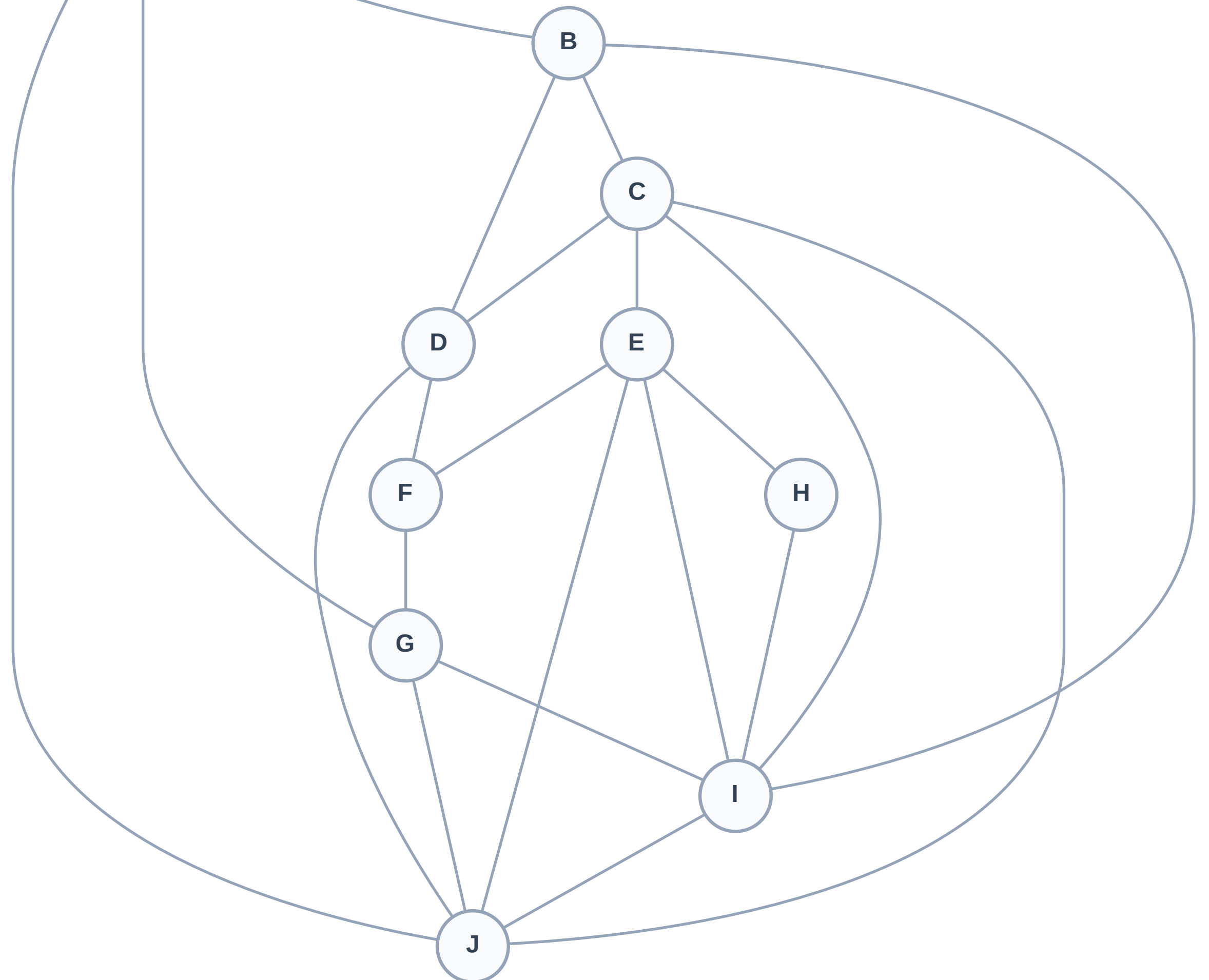
Step 2: Process node A

Legend



Stack (top at right): (empty)

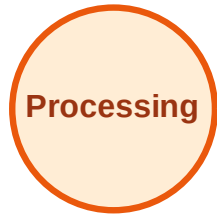
Visited: (none)



DFS Traversal

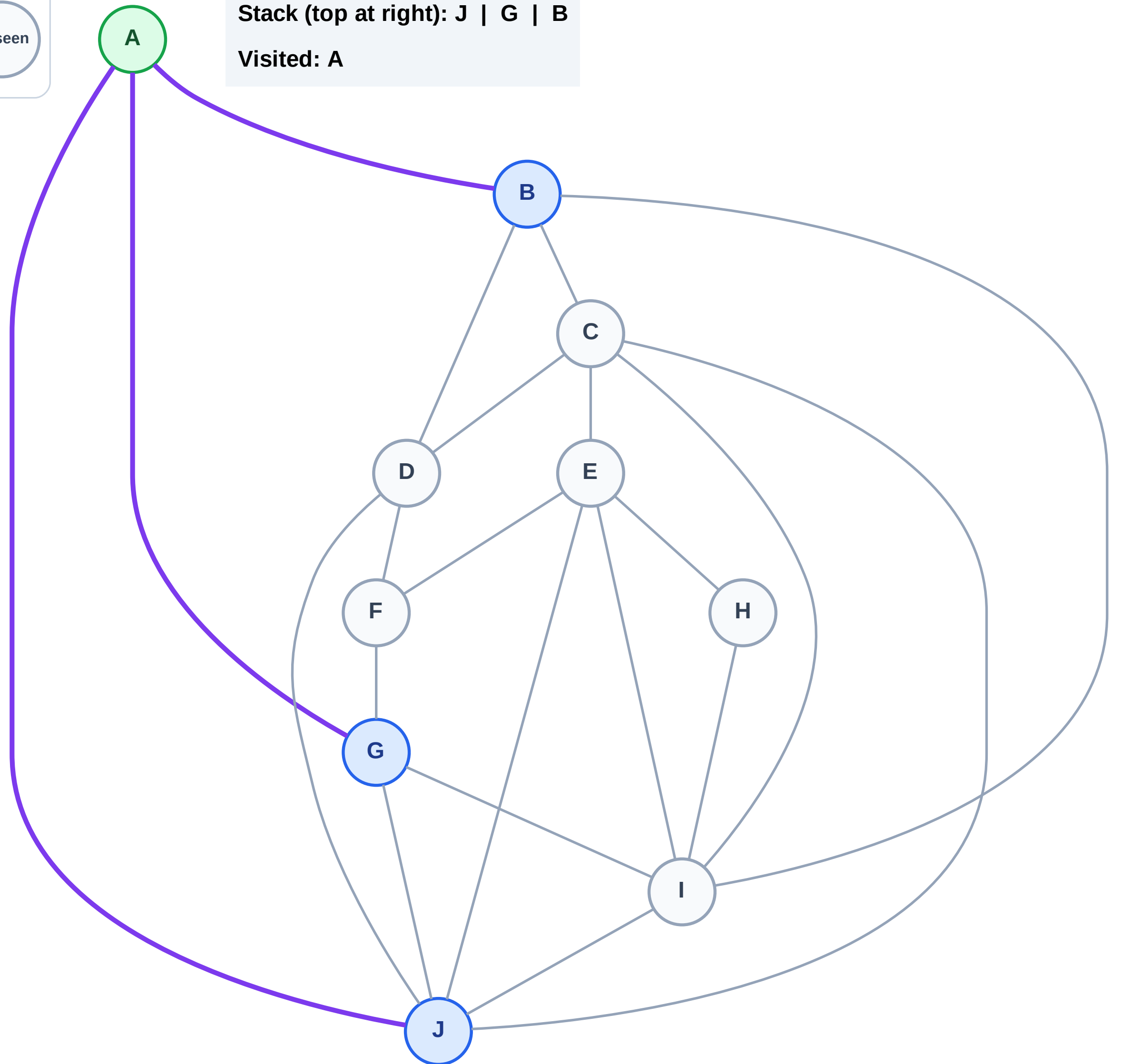
Step 3: Discover J, G, B from A

Legend



Stack (top at right): J | G | B

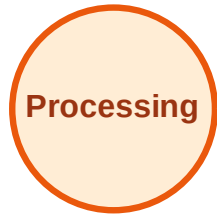
Visited: A



DFS Traversal

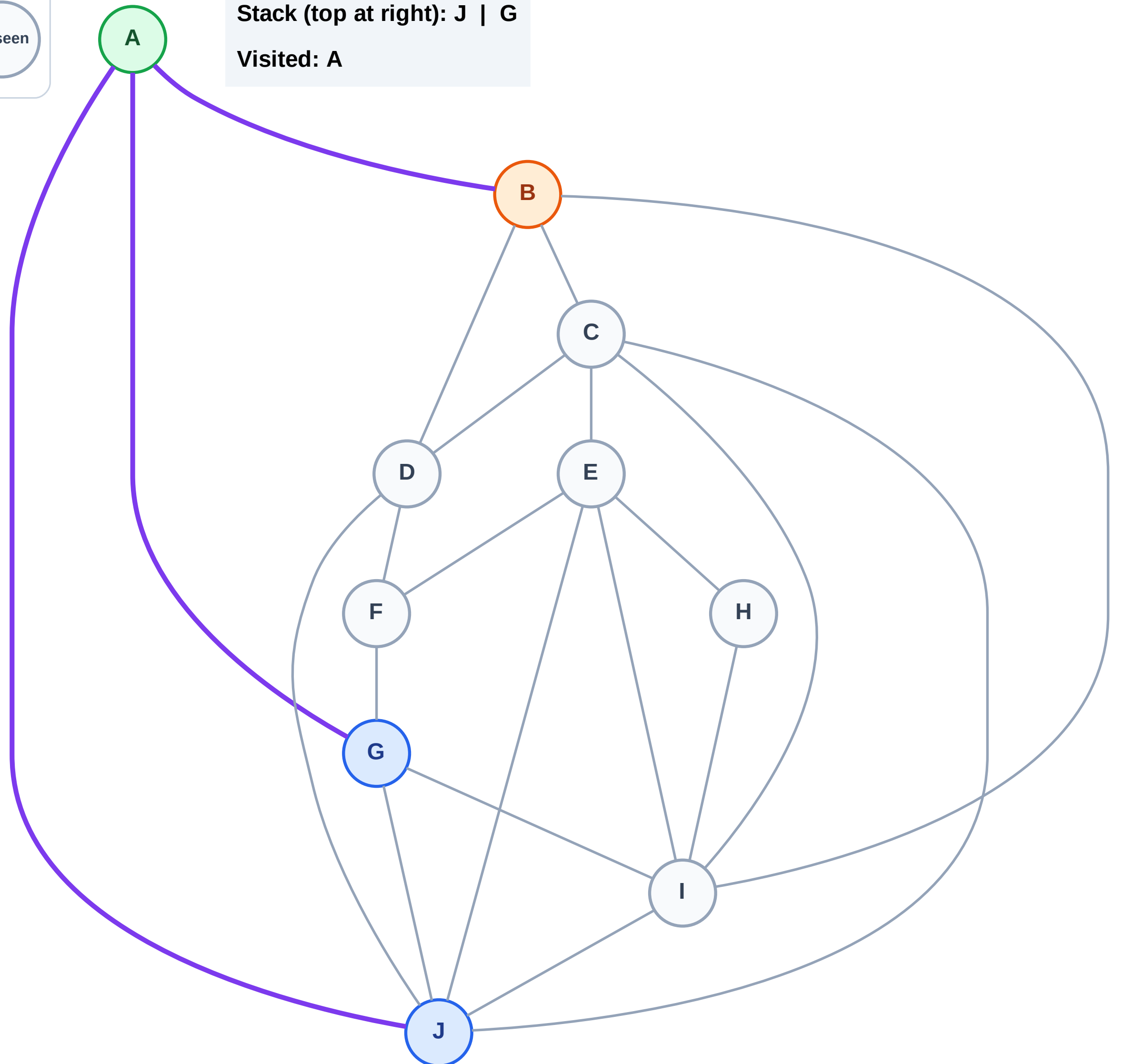
Step 4: Process node B

Legend



Stack (top at right): J | G

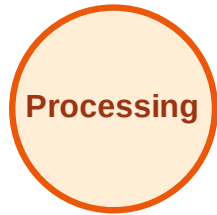
Visited: A



DFS Traversal

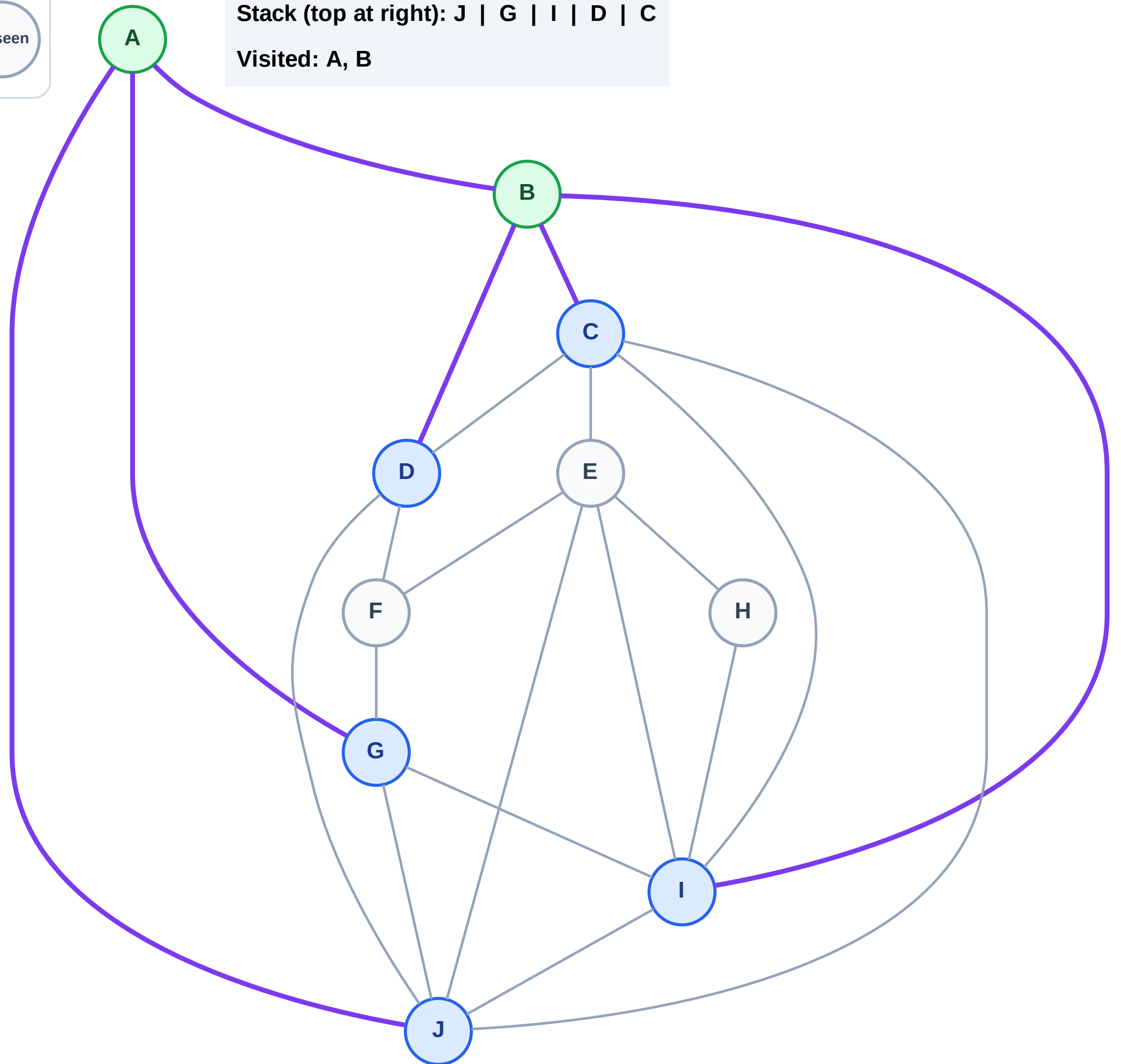
Step 5: Discover I, D, C from B

Legend



Stack (top at right): J | G | I | D | C

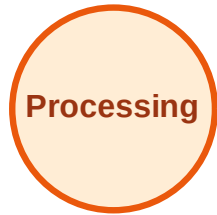
Visited: A, B



DFS Traversal

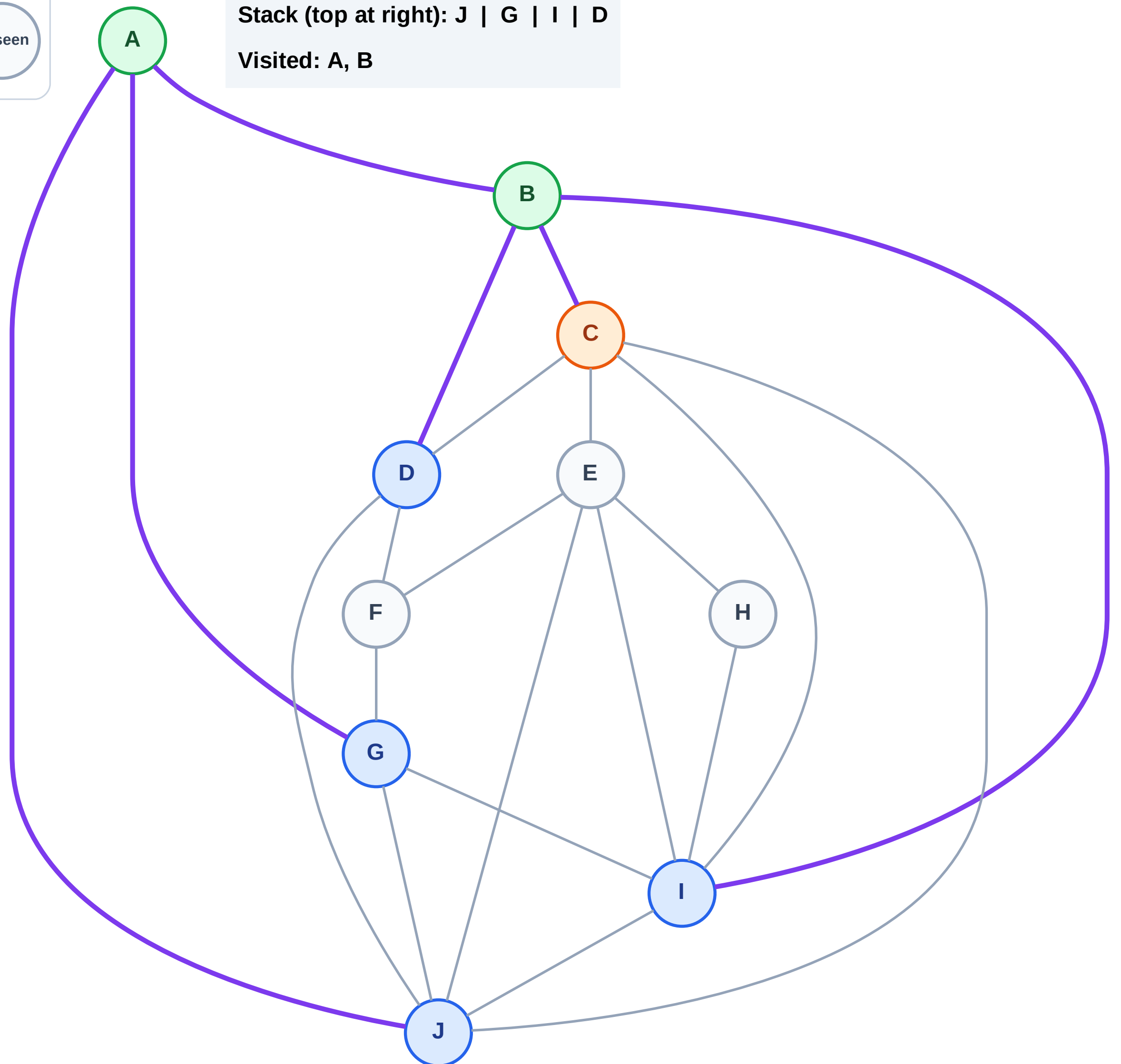
Step 6: Process node C

Legend



Stack (top at right): J | G | I | D

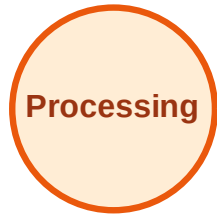
Visited: A, B



DFS Traversal

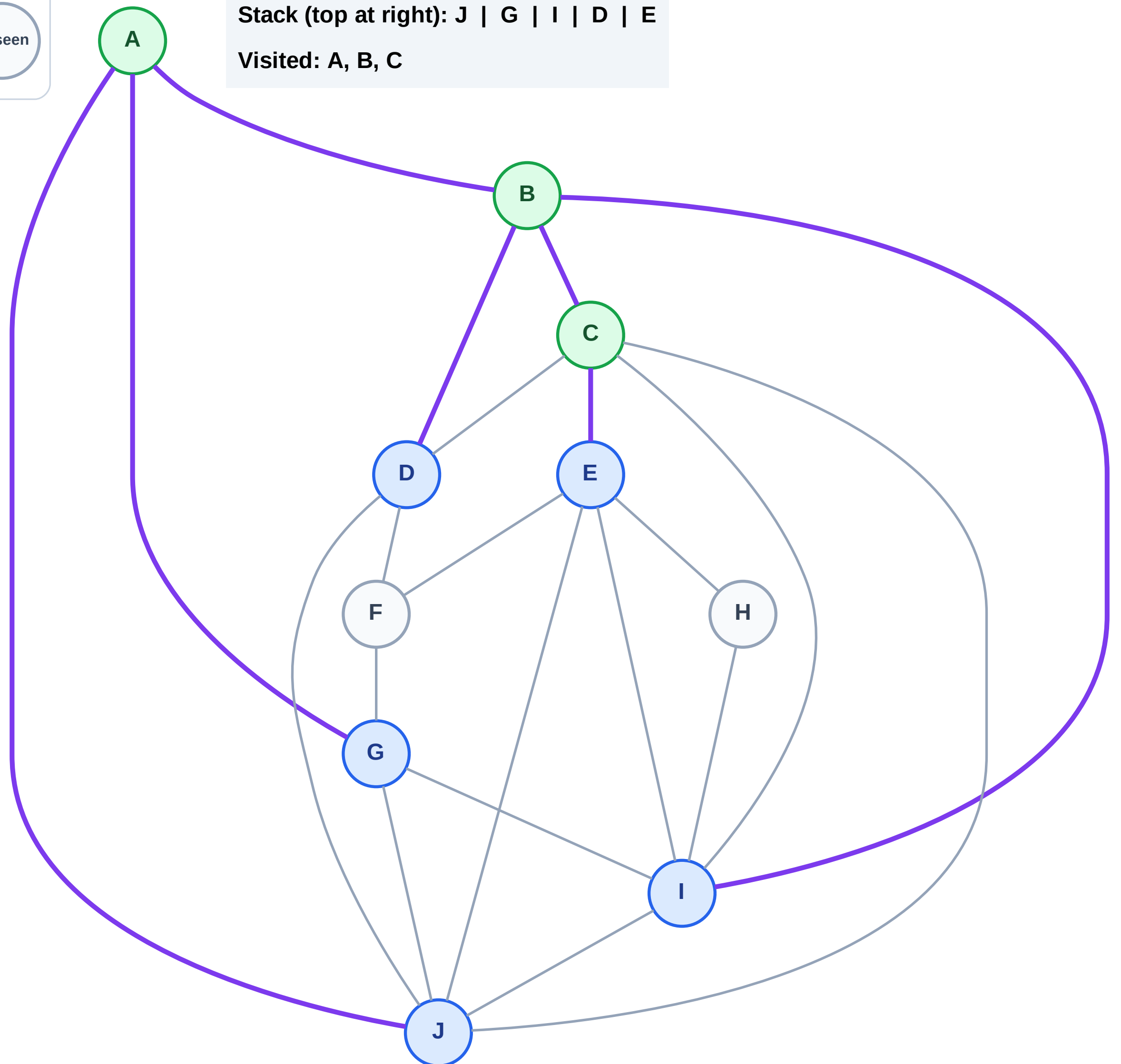
Step 7: Discover E from C

Legend



Stack (top at right): J | G | I | D | E

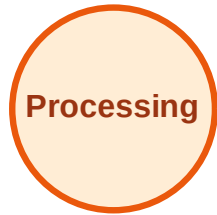
Visited: A, B, C



DFS Traversal

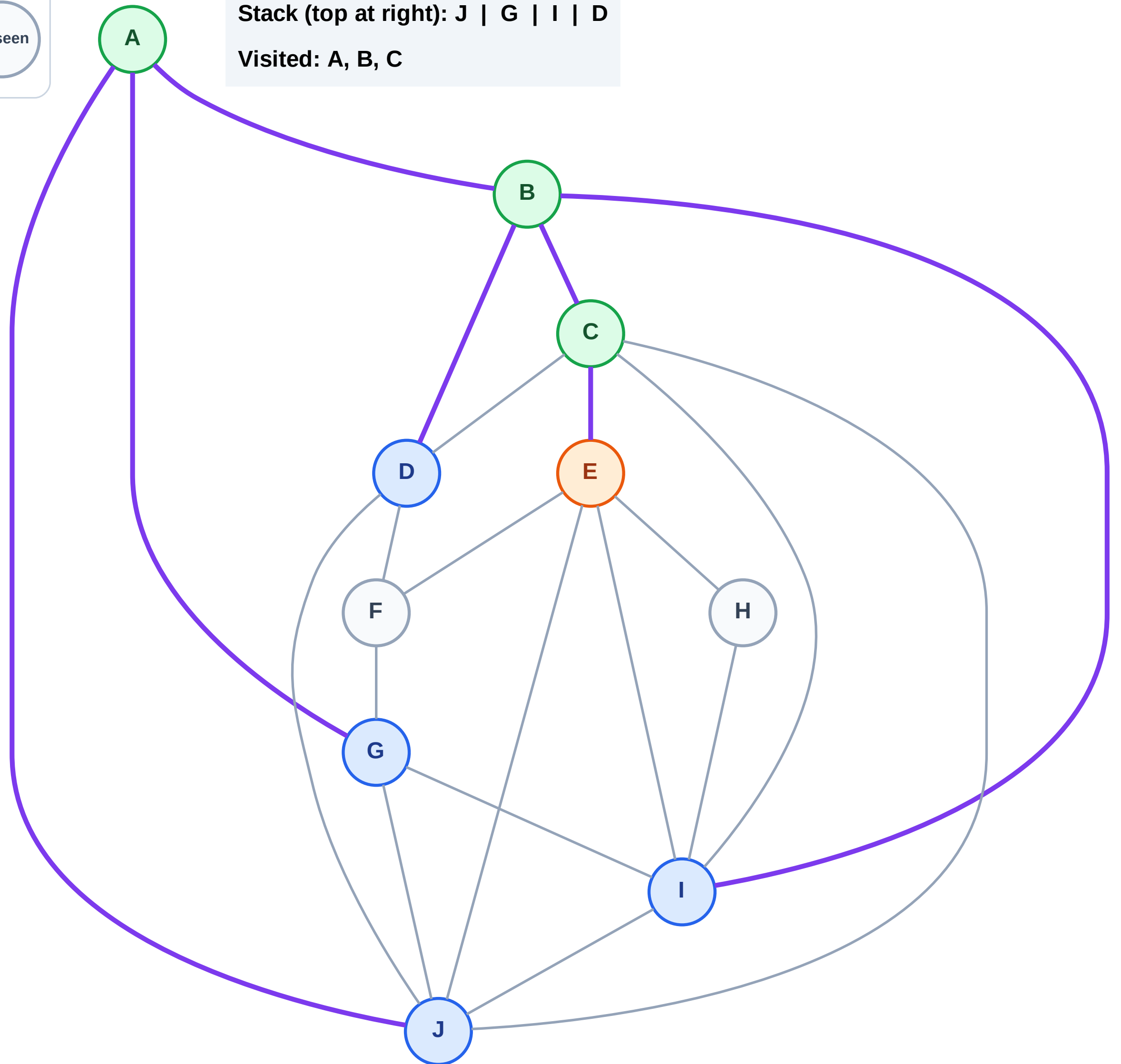
Step 8: Process node E

Legend



Stack (top at right): J | G | I | D

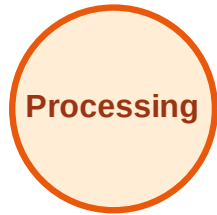
Visited: A, B, C



DFS Traversal

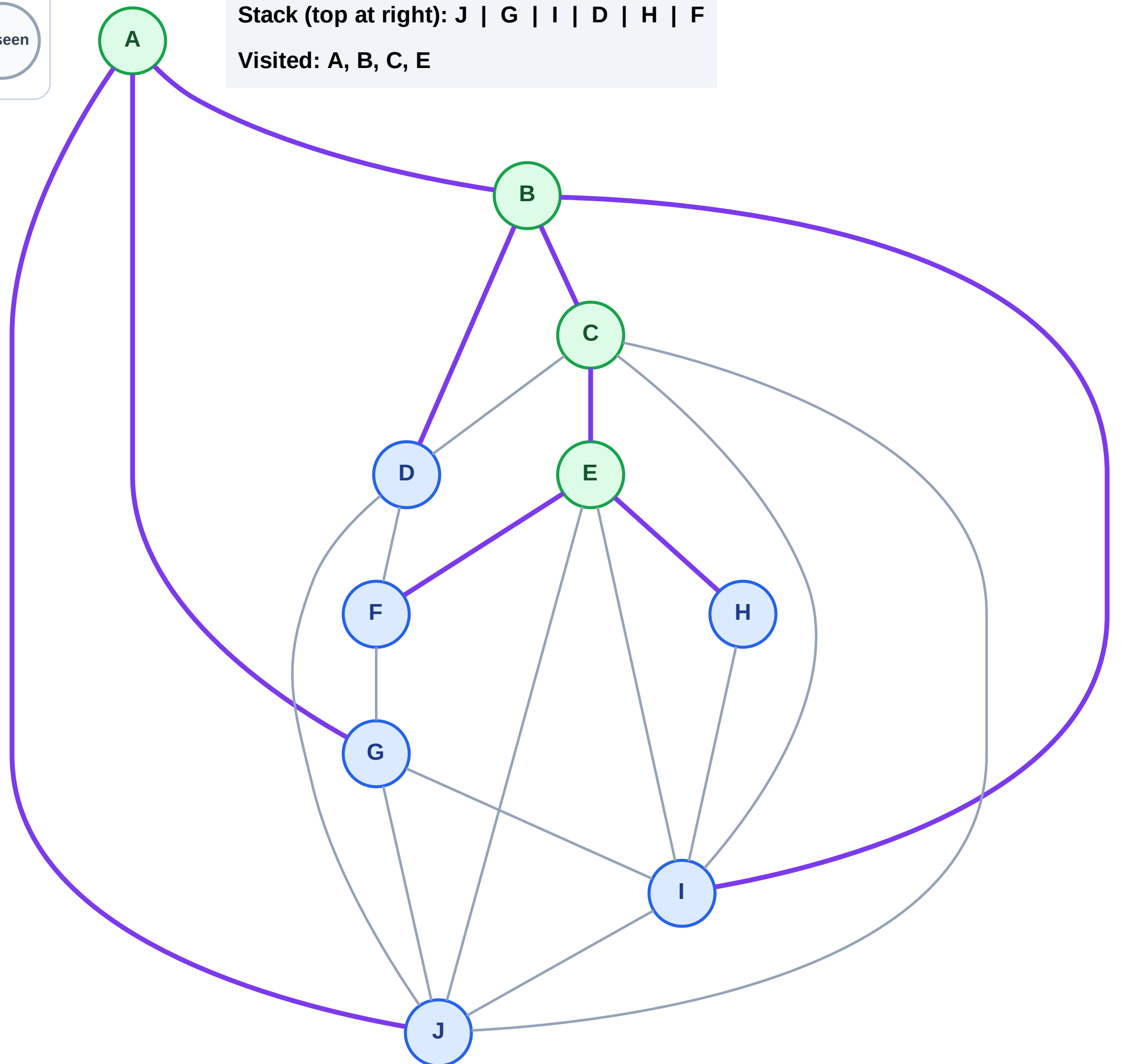
Step 9: Discover H, F from E

Legend



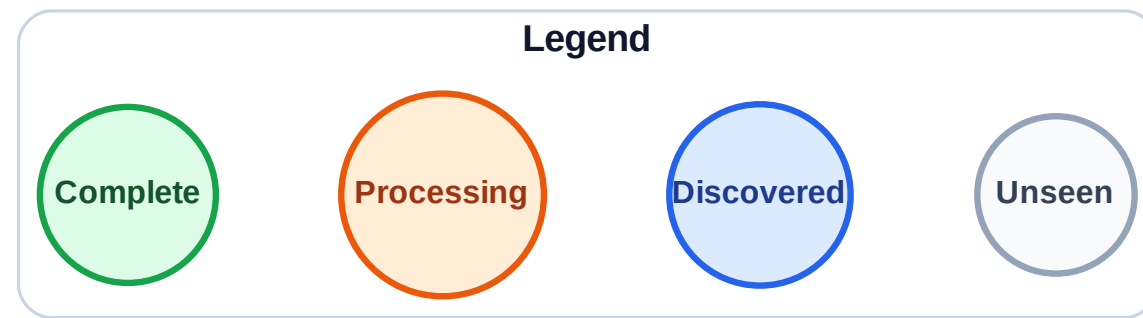
Stack (top at right): J | G | I | D | H | F

Visited: A, B, C, E



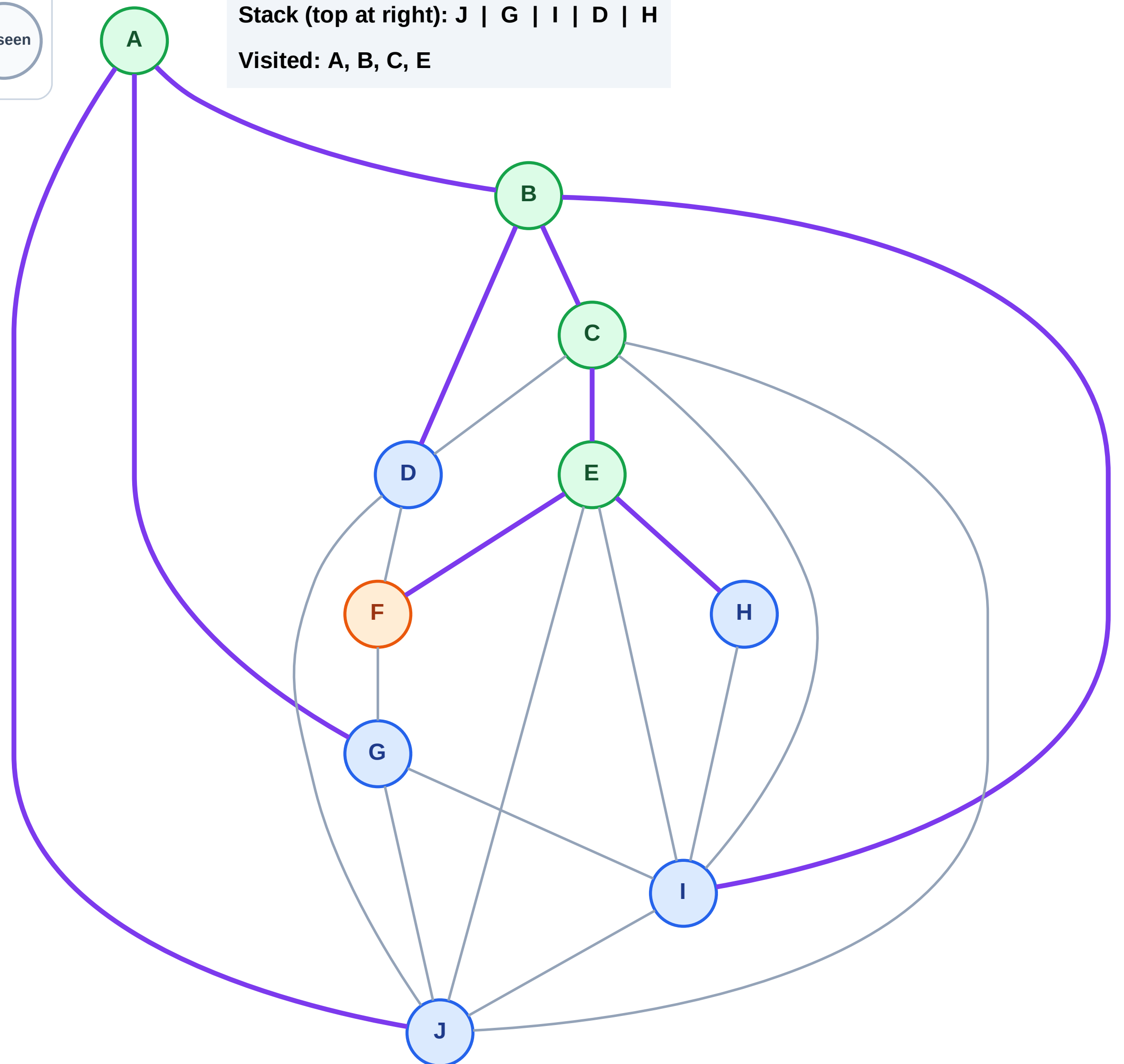
DFS Traversal

Step 10: Process node F



Stack (top at right): J | G | I | D | H

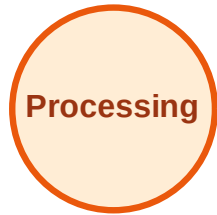
Visited: A, B, C, E



DFS Traversal

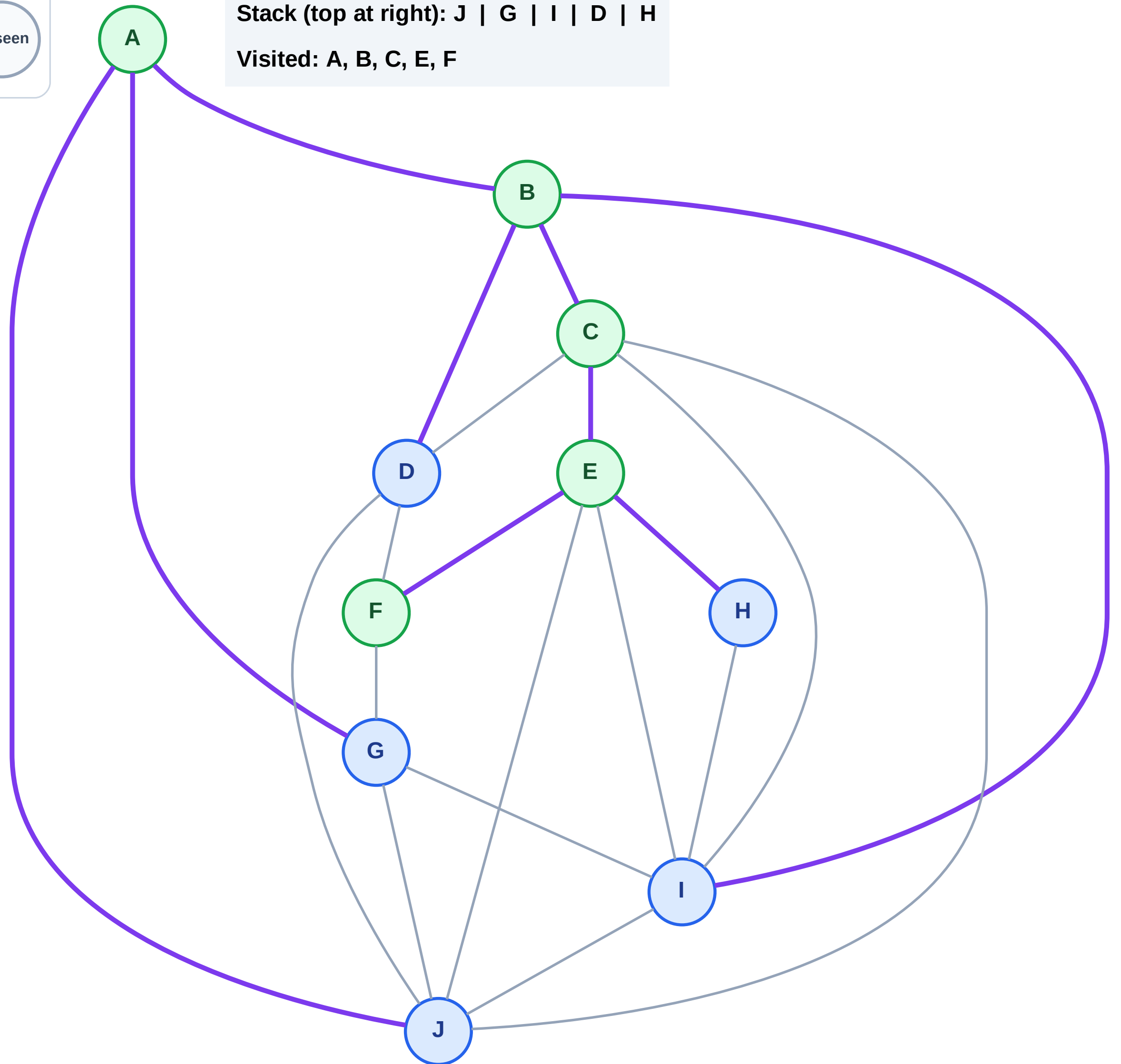
Step 11: No new neighbors from F

Legend



Stack (top at right): J | G | I | D | H

Visited: A, B, C, E, F



DFS Traversal

Step 12: Process node H

Legend

Complete

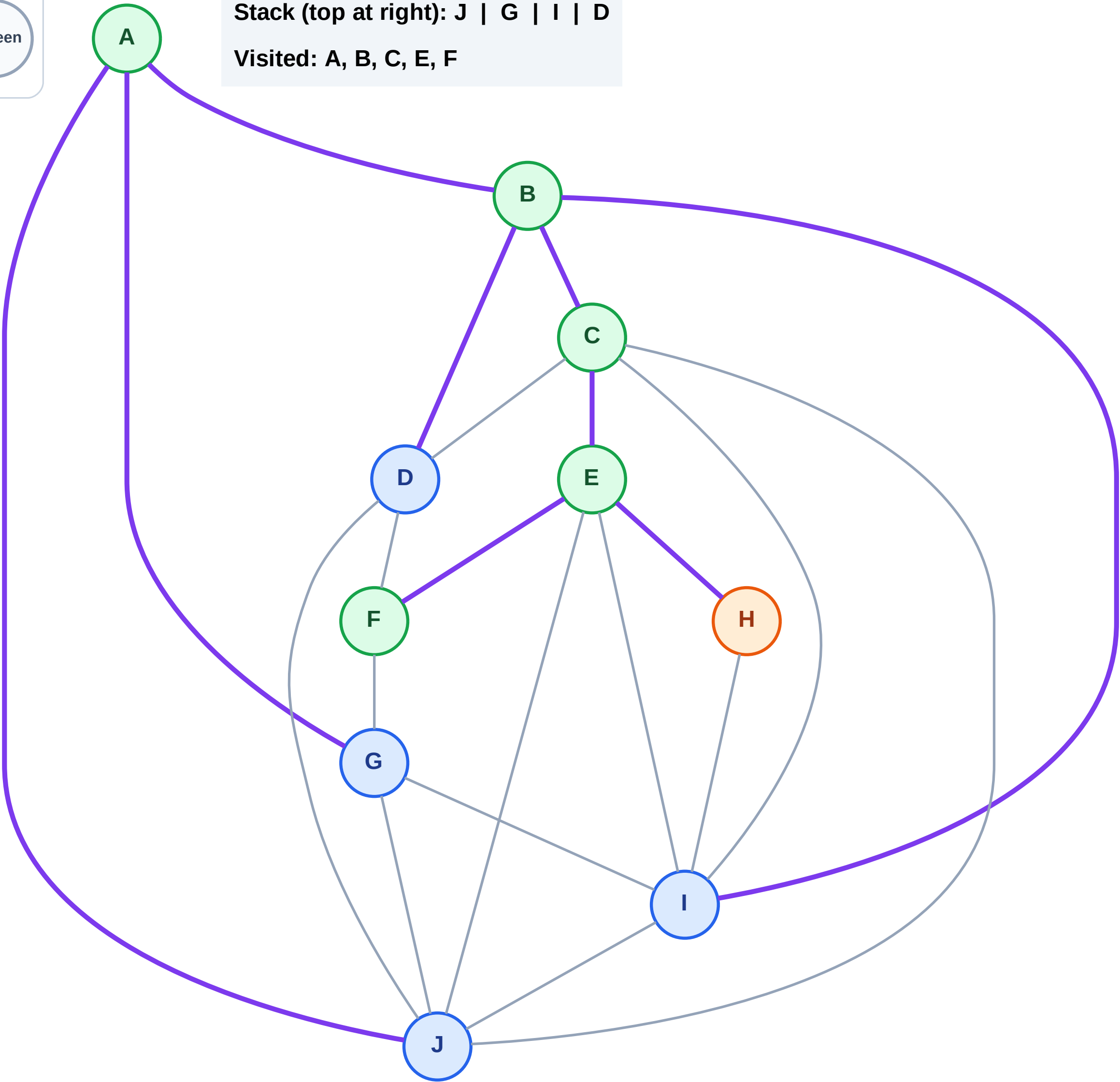
Processing

Discovered

Unseen

Stack (top at right): J | G | I | D

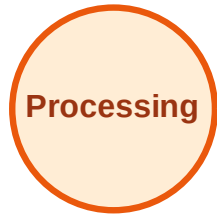
Visited: A, B, C, E, F



DFS Traversal

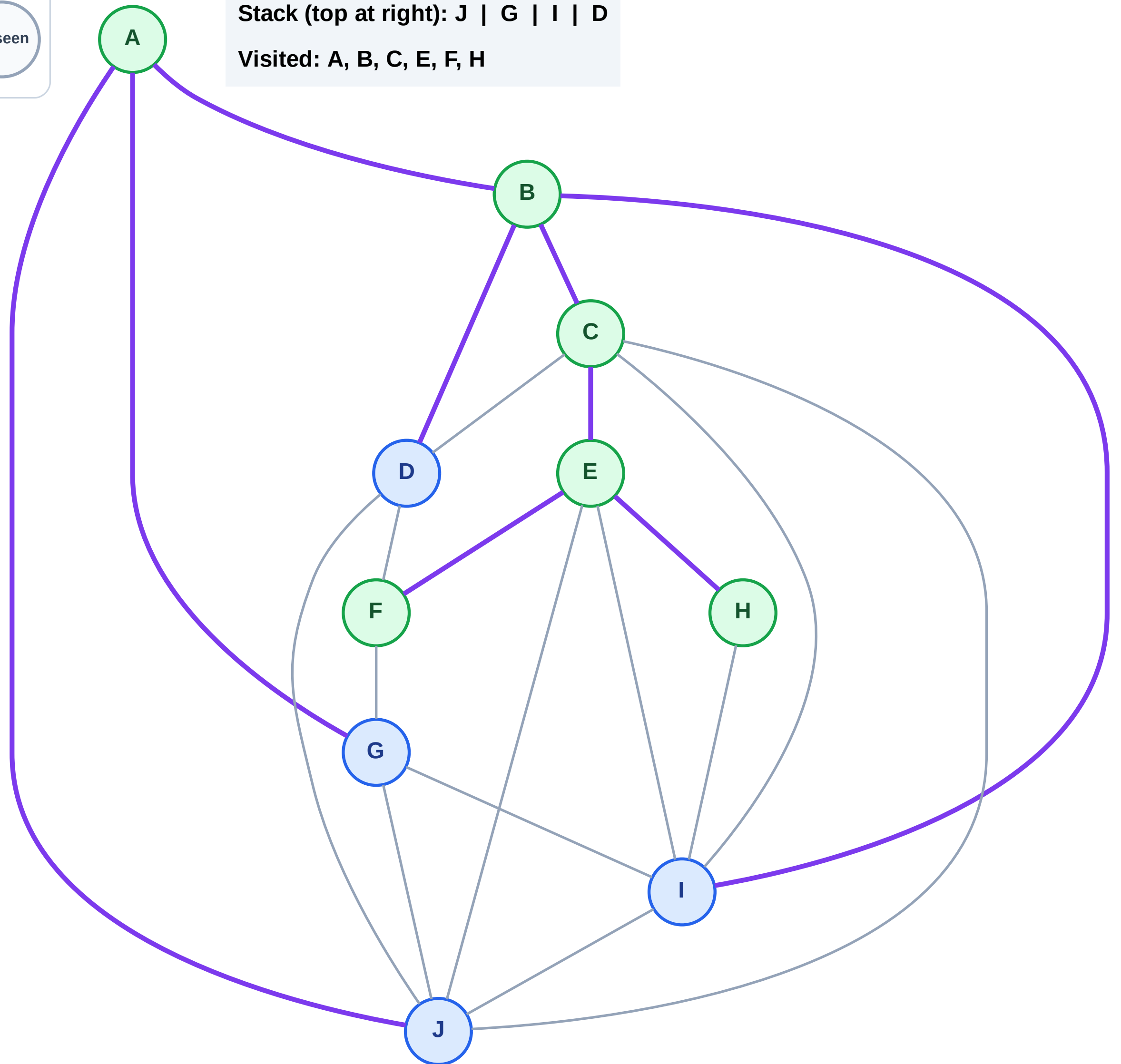
Step 13: No new neighbors from H

Legend



Stack (top at right): J | G | I | D

Visited: A, B, C, E, F, H



DFS Traversal

Step 14: Process node D

Legend

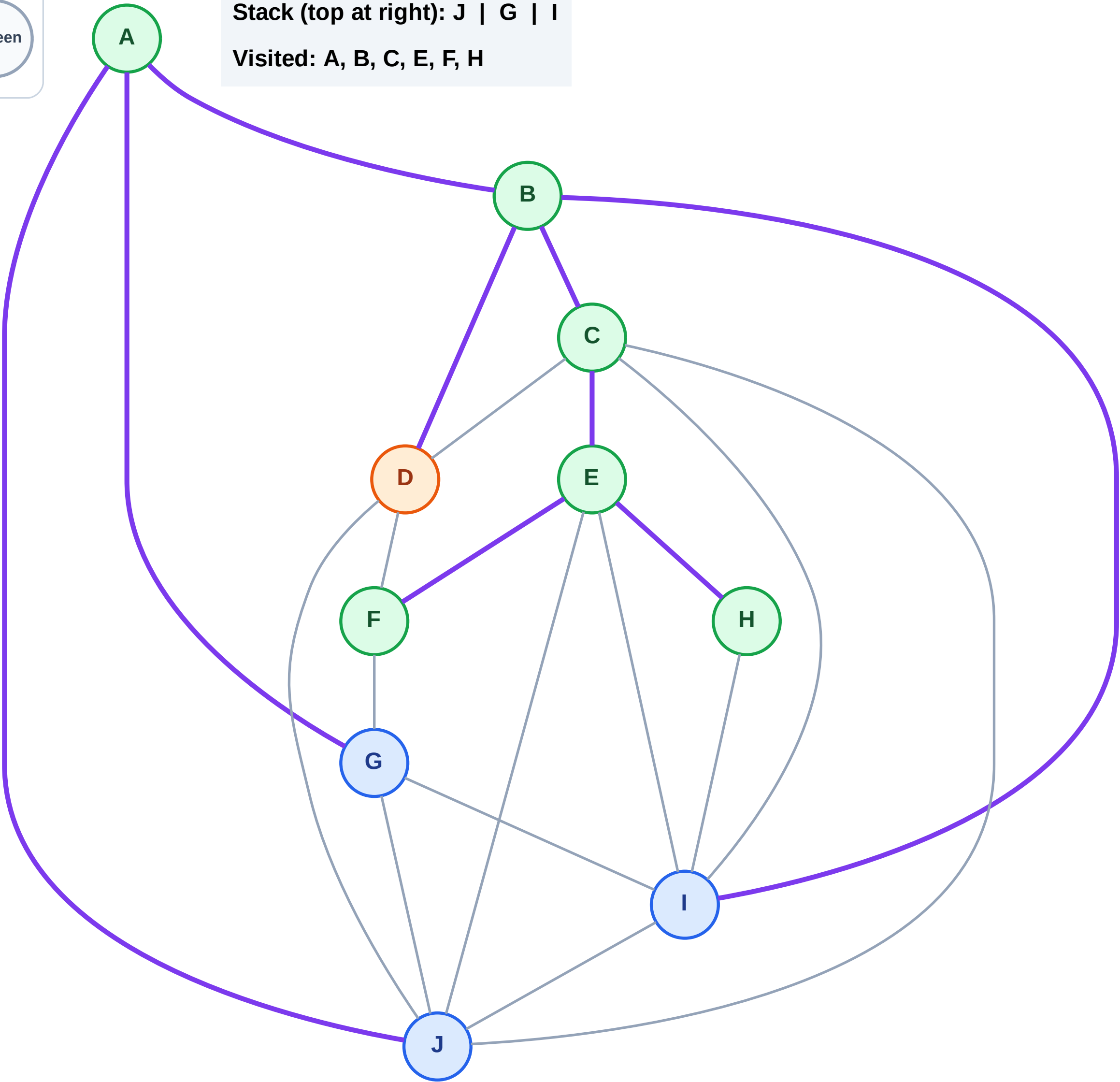
Complete

Processing

Discovered

Unseen

Stack (top at right): J | G | I
Visited: A, B, C, E, F, H



DFS Traversal

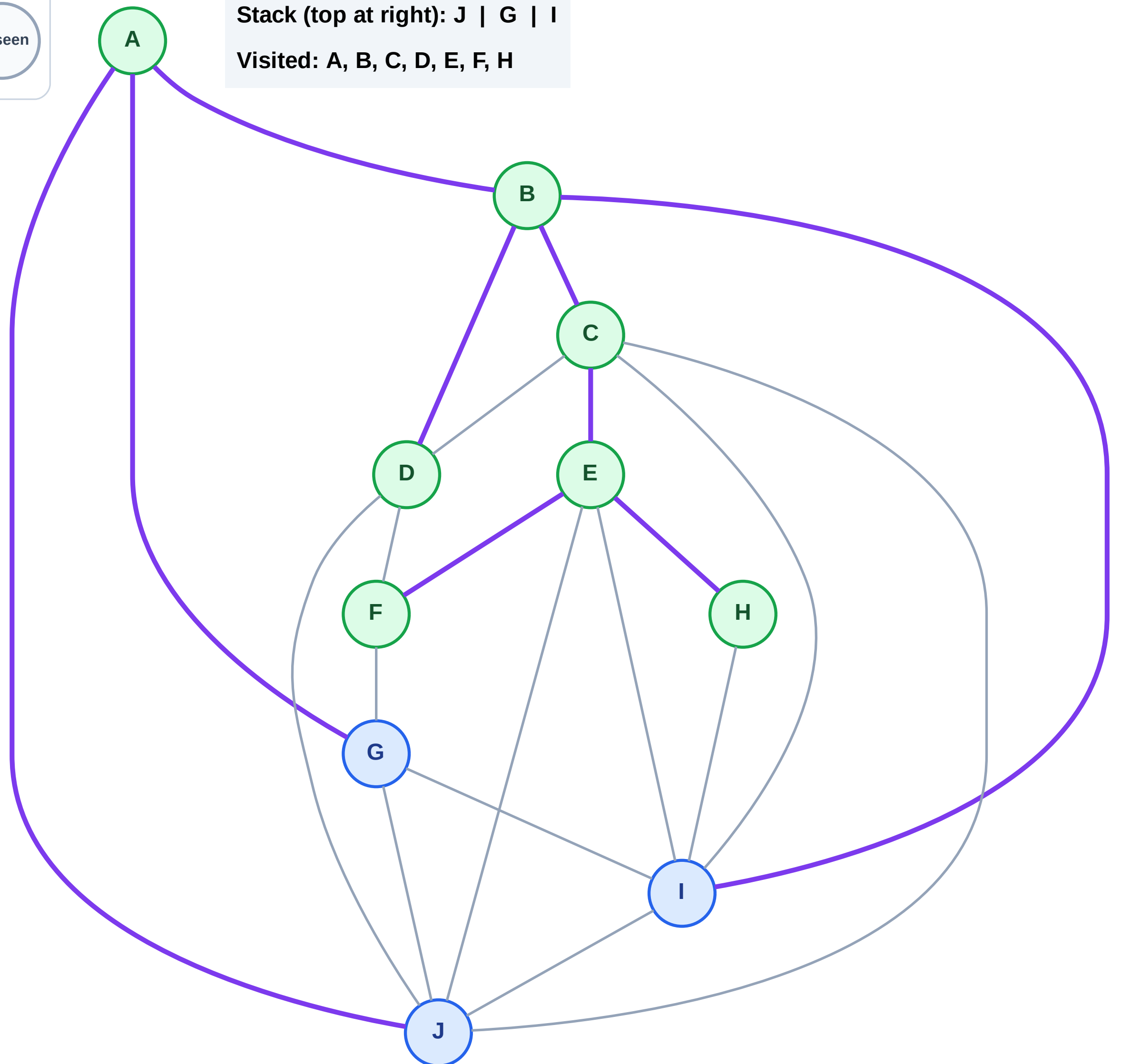
Step 15: No new neighbors from D

Legend



Stack (top at right): J | G | I

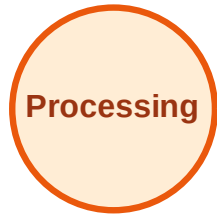
Visited: A, B, C, D, E, F, H



DFS Traversal

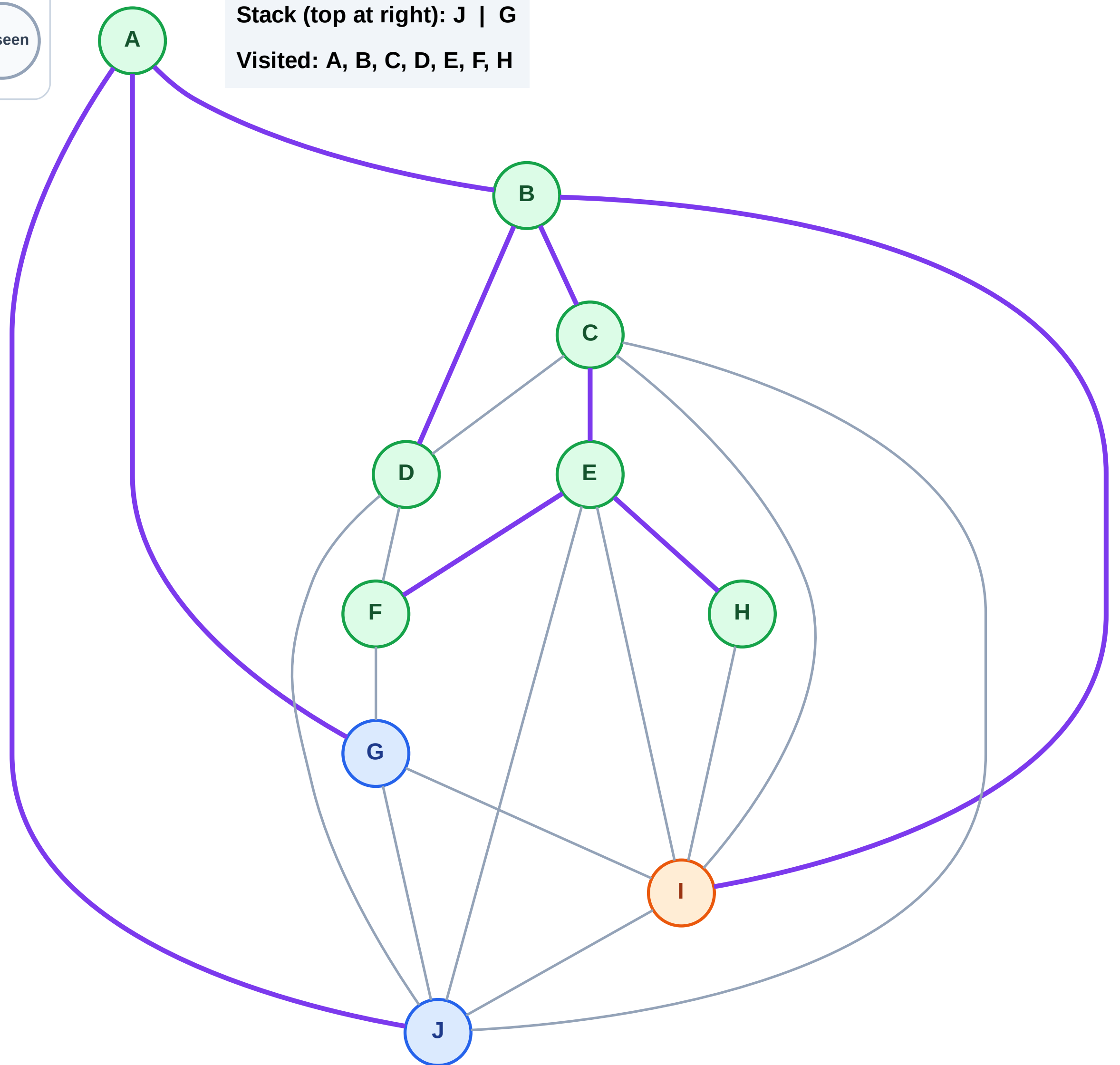
Step 16: Process node I

Legend



Stack (top at right): J | G

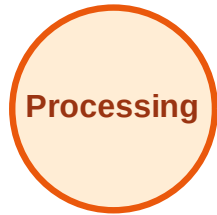
Visited: A, B, C, D, E, F, H



DFS Traversal

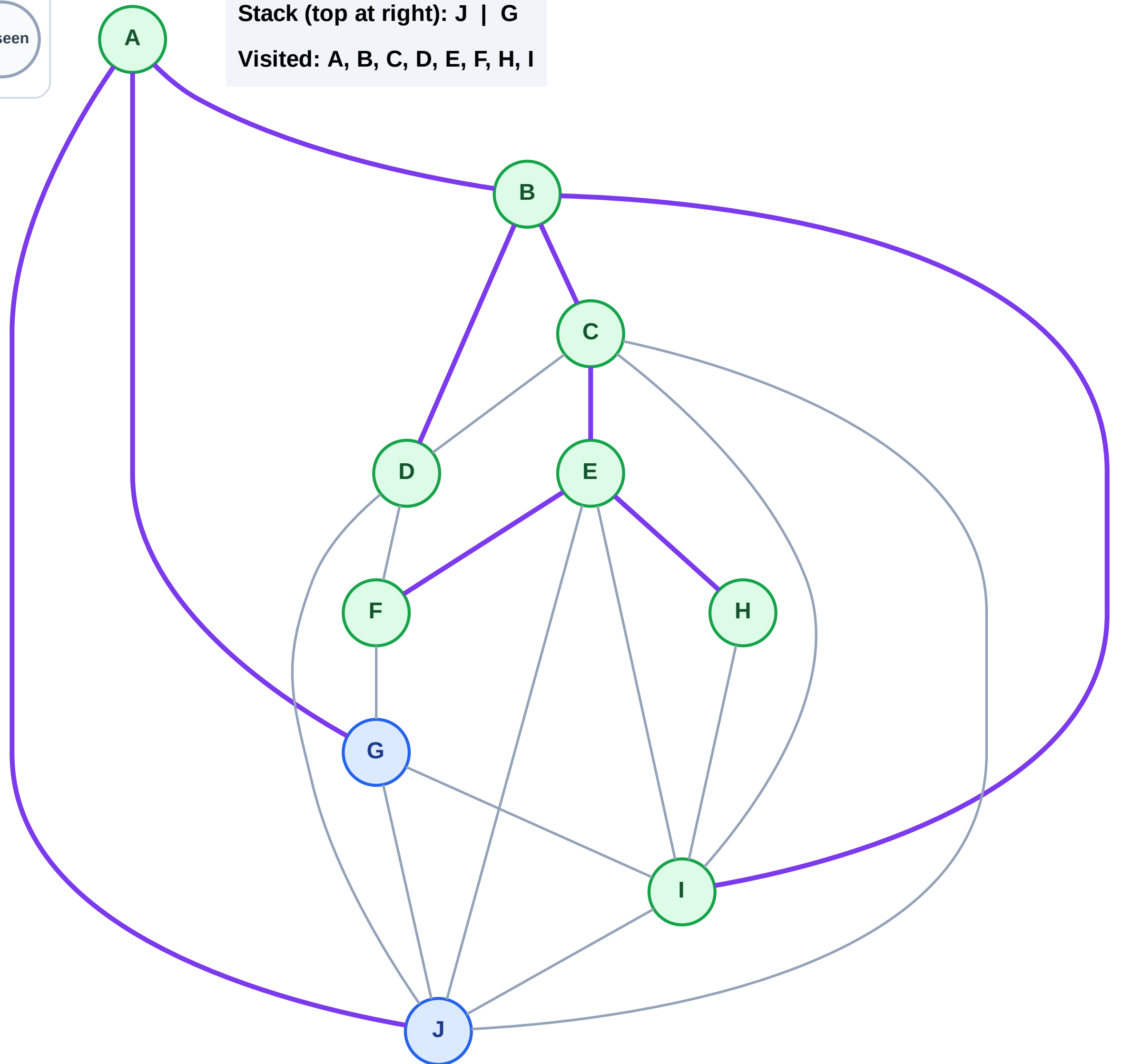
Step 17: No new neighbors from I

Legend



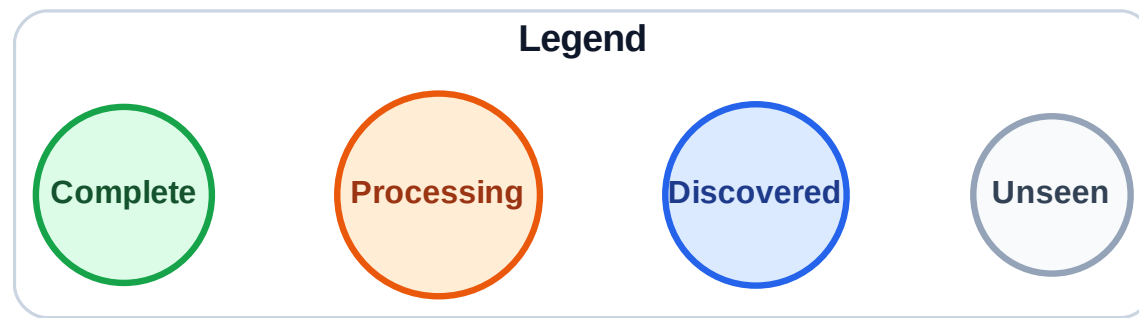
Stack (top at right): J | G

Visited: A, B, C, D, E, F, H, I

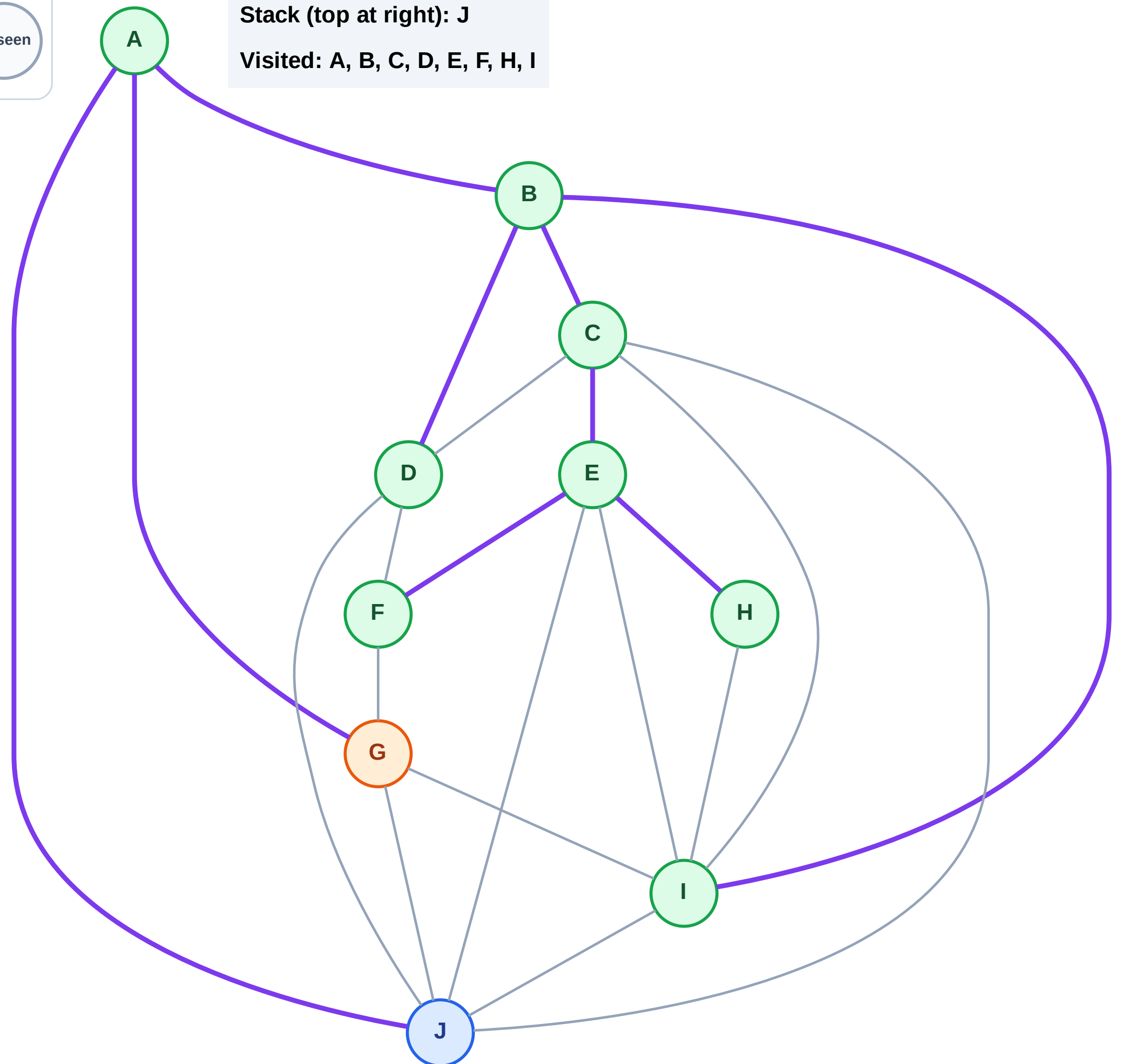


DFS Traversal

Step 18: Process node G



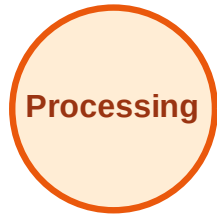
Stack (top at right): J
Visited: A, B, C, D, E, F, H, I



DFS Traversal

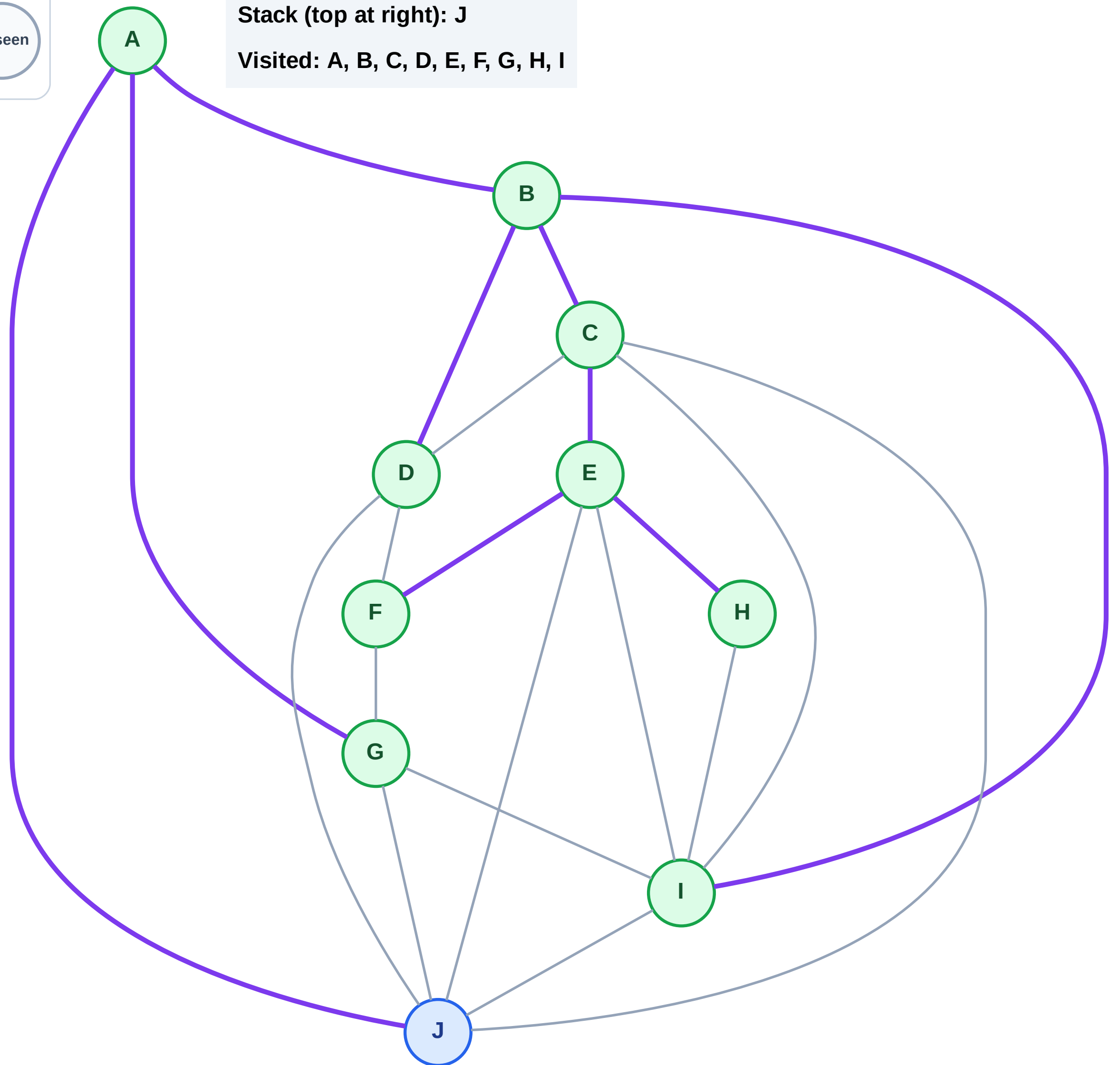
Step 19: No new neighbors from G

Legend



Stack (top at right): J

Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

Step 20: Process node J

Legend

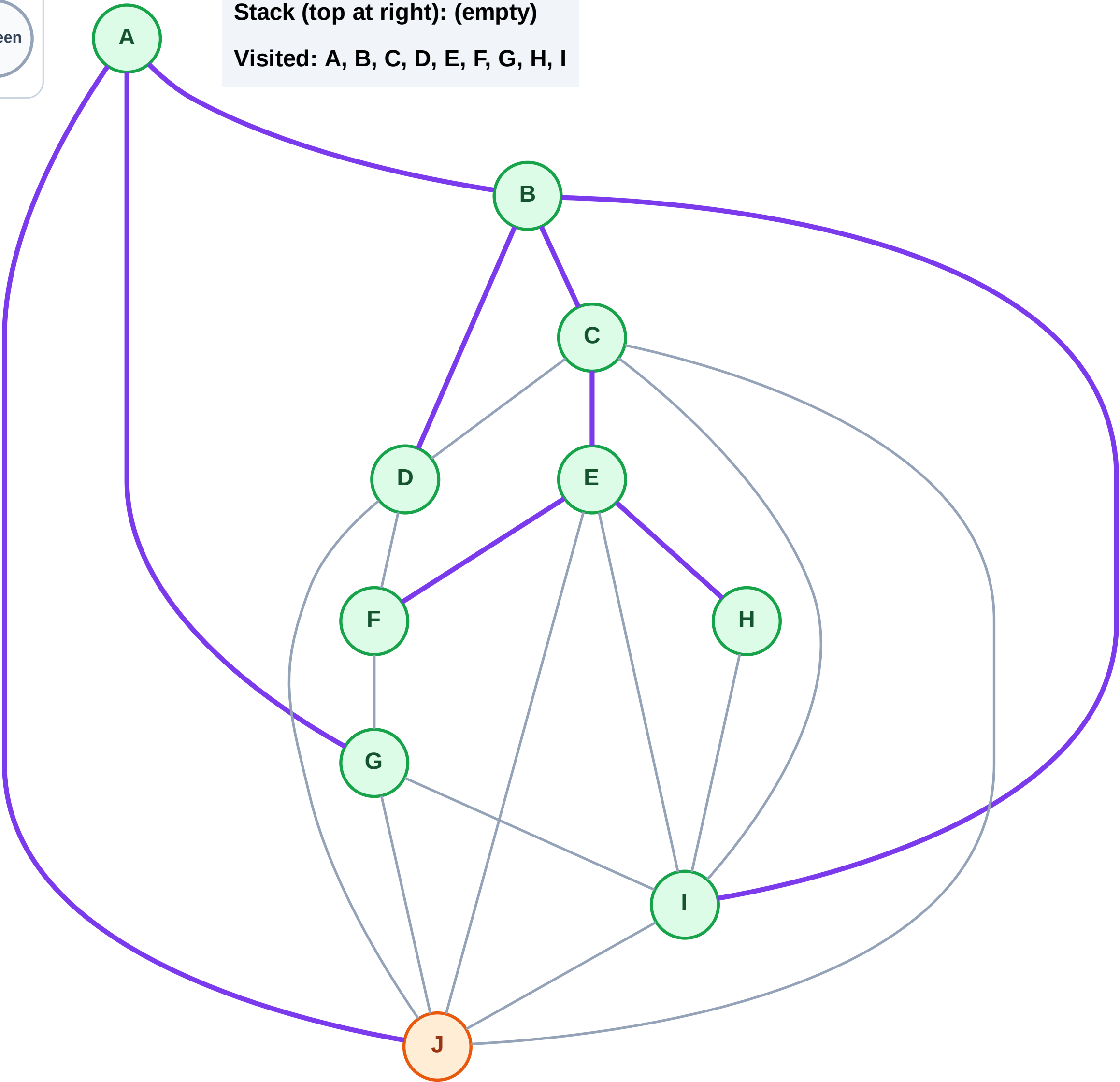
Complete

Processing

Discovered

Unseen

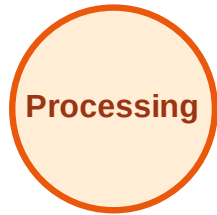
Stack (top at right): (empty)
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

Step 21: No new neighbors from J

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

